

The Sovereign Default Risk of Giant Oil Discoveries*

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Abstract

Sovereign spreads significantly increase following giant oil discoveries. This is puzzling to canonical sovereign default models, in which spreads decrease when future income increases. I develop a model with oil discoveries where financial frictions generate an endogenous default penalty. Sectoral differences in technology imply different exposure to the default penalty. This exposure is summarized by a sufficient statistic that can be computed from input-output tables which shows that the oil sector is relatively less exposed to the default penalty in a set of oil-producing economies. This allows the model to successfully generate the puzzling response while matching business cycle regularities.

Keywords: Sovereign risk, oil discoveries, news shocks.

JEL Codes: F34, F41.

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1 Introduction

Between 1970 and 2012, sixty-four countries discovered at least one giant oil field, and fourteen of these countries had a default episode in the following ten years.¹ This paper studies how these discoveries impact sovereign risk and debt accumulation.

I build on the work of [Arezki, Ramey, and Sheng \(2017\)](#), who document the effects of giant oil field discoveries on macroeconomic aggregates. These discoveries can be interpreted as news of higher future income since there is an average gap of 5.4 years between a discovery and when production on the field starts. Following a discovery, countries experience a current account deficit which reverts around the time when the field becomes productive. This is consistent with the permanent income hypothesis, since an increase in GDP is only observed after production on the field starts. Following their methodology, I estimate the effect of discoveries on sovereign spreads and I find that it is large and positive: spreads increase by up to 1.2 percentage points following a discovery of median size. In addition, I document that government consumption and borrowing also increase following a discovery, while private consumption remains mostly unchanged.

These joint responses of spreads and government borrowing are puzzling from the point of view of standard sovereign default theory for two reasons. First, government borrowing increases despite the higher cost. In the data—and in standard models—government borrowing increases when spreads are low, not high. Second, spreads increase following oil discoveries, which are large and positive shocks to income. In the data, spreads are countercyclical and default events happen when output is relatively low. A key assumption that standard models make to replicate these co-movements is an output cost of default that is asymmetrically increasing with income (see the discussion in [Arellano \(2008\)](#) and the literature thereafter). If all output were subject to the same increasing costs from an eventual default then news of an oil discovery would reduce default incentives and spreads. In fact, they increase in the data.

This evidence suggests that income from the oil sector may have different effects on default incentives than income from other sources, which stresses the importance of a detailed theory of the real cost of default. There is ample literature arguing that disruptions in financial markets are at the core of GDP contractions and productivity losses during default episodes and sovereign debt

¹A giant oil field contains at least 500 million barrels of ultimately recoverable oil. “Ultimately recoverable reserves” is an estimate (at the time of the discovery) of the total amount of oil that could be recovered from a field.

crises (see [Reinhart and Rogoff \(2011\)](#), [Bocola \(2016\)](#), [Laeven and Valencia \(2018\)](#), [Arellano, Bai, and Bocola \(2017\)](#)). I develop a quantitative sovereign default model with oil discoveries, firms that face financial frictions, and production in different sectors that reconciles existing theory with the responses of spreads to oil discoveries from the data. Default in the model causes an endogenous efficiency loss as the economy gets excluded from international credit markets. This is because firms in the model require working capital to purchase a subset of imported inputs, so when access to credit markets is disrupted these inputs are replaced by domestic and other imported imperfect substitutes. This core mechanism of the model builds on the production structure and financial frictions developed by [Mendoza and Yue \(2012\)](#), but allows for technology and working-capital needs to differ across sectors. These differences jointly determine how exposed each sector is to the efficiency loss in default, and are the core of the mechanism that explains the response of spreads to discoveries. If the oil sector is relatively less exposed, then oil discoveries lower the utility loss of defaulting in any given state. This is because a discovery raises the relative size of the oil sector, so a larger share of consumption is financed with a source of income that is less affected by a potential default, which increases default incentives and, with them, spreads. I show that this mechanism is summarized by a sufficient statistic that can be computed from an input-output table.

I first calibrate the model to Mexico, which is a typical small-open economy widely studied in the sovereign debt and emerging markets literature. The data from Mexico's input-output table feature an oil sector that uses fewer intermediate inputs than the rest of the economy and that imports a smaller share of them. Another advantage of using Mexico as a calibration benchmark is that oil extraction is mostly done by a large government-run firm, which was a constitutionally sanctioned monopoly between 1938 and 2014. This allows me to use financial statements from this firm to directly discipline the oil sector's reliance on working capital, which also turns out to be smaller than in the rest of the economy.

Overall, the Mexican data unambiguously imply an oil sector that is relatively less exposed to the endogenous cost of default. Following an oil discovery, default incentives and spreads increase. The large size of oil discoveries drives an increase in borrowing to front-load consumption, despite its higher costs. This is how the model successfully generates the puzzling joint responses of spreads and other macro aggregates to giant oil discoveries while also matching other business cycle regularities, the latter driven by standard productivity shocks. I then compute the sufficient

statistic described above for a cross-section of oil-producing economies and find that the condition under which a discovery worsens borrowing terms holds for most of them, so the main result is not particular to Mexico. Moreover, I show that the result survives even when the model is calibrated to the economy in the sample whose sectoral differences are furthest from Mexico's. Reversing the sign requires sectoral differences that are jointly more extreme than any observed in the data.

The oil sector in the model is operated by a government-owned company, to be consistent with the benchmark calibration to Mexico. The direction of the results, however, does not depend on this assumption: spreads still increase in an alternative model with the oil sector run by private firms. The key mechanism relies on the differences in technology and financing needs that shape the sector's exposure to the real costs of default, not on who operates it.

Finally, I explore the welfare implications of oil discoveries under different scenarios. In the benchmark model, average gains are 7.6 percent, expressed in consumption equivalent units. These are large, but small compared to the average increase in government consumption of 20 percent. Interestingly, the gains are smaller if domestic private firms operate the oil sector. This is because the government has access to a smaller fraction of the windfall, so the quantity of resources that can be front-loaded is limited (only the government has access to international borrowing in the model). Selling the oil field to foreigners also yields smaller gains, because the government values the rents it gives up by more than the risk-free price foreigners pay for them. It values them by more precisely because they are immune to the costs of defaulting, which is the same property that makes spreads increase after a discovery.

Related literature.—This paper contributes to the quantitative sovereign default literature following [Aguiar and Gopinath \(2006\)](#) and [Arellano \(2008\)](#), which extend the approach developed by [Eaton and Gersovitz \(1981\)](#). They introduce models that feature countercyclicality of net exports and interest rates, which are consistent with the data from emerging markets. This paper presents evidence of a relatively rare, but large, income shock that has an opposite relationship with spreads. The main contribution is to reconcile standard sovereign default theory with this finding. Importantly, I show that this can be done by building on the canonical model of the real cost of default in general equilibrium developed by [Mendoza and Yue \(2012\)](#).

This paper is closely related to [Hamann, Mendez-Vizcaino, Mendoza, and Restrepo-Echavarria \(2023\)](#), who analyze how oil-reserves management affects sovereign risk. They study oil reserves

as an implicit form of capital that the government controls, so oil interacts with sovereign risk through how foreign investors price the strategic management of extraction. In contrast, the government controls neither the size nor the timing of the discoveries that I study, and oil fields take several years to produce after their discovery. The key theoretical difference between the two papers is the mechanism through which the potential to produce oil increases spreads. In their model, spreads rise with the accumulation of oil reserves because accumulating them is a government choice that improves the value of defaulting. In my model, spreads rise following an exogenous oil discovery because of the change in the composition of output that it implies: a discovery increases the size of a sector that is less exposed to the cost of default. In both cases more oil-production potential raises the value of defaulting, but it raises the value of repaying as well, so the direction of spreads depends on which of the two rises more. What tilts the comparison in their environment is a choice that the government makes, whereas in mine it is the difference in technology and financial frictions across sectors. The empirical work differs as well. By definition, proved reserves do not immediately incorporate giant oil discoveries, and the size of their year-to-year changes is much smaller.² Discoveries are also rare, and their timing is exogenous enough to allow for a quasi-natural experiment approach to identify their effect. My focus on giant discoveries is more appropriate to measure the mechanism in this paper because they constitute large and discrete changes in the expected future composition of output, which arrive years before the field produces and before any extraction decision. On the other hand, marginal revisions to proved reserves mix news about future oil with the government's response to it. The two papers therefore answer different questions: theirs speaks to how the management of reserves responds to oil prices and to sovereign risk, which my model cannot address because a field's production profile is exogenous, while mine measures the response of spreads to oil news that the government did not choose and identifies the sectoral composition of the default cost behind it.

In a related paper, [Drechsel and Tenreyro \(2018\)](#) study the role of fluctuations in international commodity prices on business cycles in commodity exporters. They develop a small-open economy model in which these fluctuations affect both the competitiveness of the economy and its borrowing terms, as higher commodity prices are associated with lower spreads between the country's borrowing rate and world interest rates. The nature of the shocks they study is different from

²See Appendix [A](#) for a detailed discussion of the differences between these measures.

the oil discoveries that I study in two dimensions. First, commodity prices are driven by high-frequency shocks, while oil discoveries are relatively rare. Second, the sign of the correlation between commodity prices and sovereign spreads is negative, while it is positive for that between spreads and oil discoveries.

This paper also contributes to the literature that studies the role of news as drivers of business cycles (see [Beaudry and Portier \(2014\)](#), [Jaimovich and Rebelo \(2008\)](#), and [Arezki, Ramey, and Sheng \(2017\)](#)). The model in Section 3 builds on the work in these papers and connects it with the sovereign default literature. In a related paper, [Dvorkin, Sanchez, Sapriz, and Yurdagul \(2020\)](#) study the effects of news shocks about future productivity on sovereign default risk and also develop a sovereign default model that connects with the news-in-business-cycles literature. They find that these productivity news shocks have a larger contemporaneous impact on spreads than a comparable shock to labor productivity and study their implications for the optimal management of debt maturity. The oil discoveries that I study have similarly large effects but the opposite sign, which drives my focus on the key mechanism driving this puzzling behavior.

Layout.—Section 2 presents the empirical analysis and discusses the evidence that motivates the theoretical framework. Section 3 presents the model. Section 4 performs the quantitative analysis. Section 5 concludes.

2 Sovereign spreads and oil discoveries

In this section, I document that sovereign spreads and government borrowing in emerging economies increase following giant oil discoveries. I use a simple quantitative model of sovereign default with oil discoveries to illustrate why these responses are puzzling and potential drivers of the co-movements. Then, in the following section, I present a richer model with production that reconciles both the novel empirical findings with the business cycle regularities of emerging economies.

2.1 Data

The data on giant oil field discoveries in the world was collected by [Horn \(2014\)](#) and the Global Energy Systems research group at Uppsala University. The size of a newly discovered oil field is measured by its “ultimately recoverable reserves” (URR), which is an estimate (at the time of the

discovery) of the total volume of oil that could be recovered from a field. An oil field is considered “giant” if it contains at least 500 million barrels of URR. The data in [Horn \(2014\)](#) include all giant oil field discoveries in the world between 1960 and 2012.

To estimate the effects of discoveries on macroeconomic variables, I build on the work by [Arezki, Ramey, and Sheng \(2017\)](#), who construct a measure of the net present value (NPV) of oil discoveries relative to the GDP of the country where the discovery happened. They construct this measure as follows:³

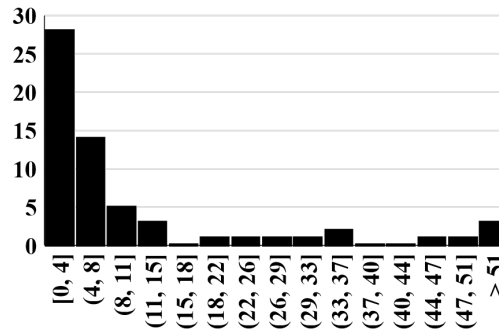
$$NPV_{i,t} = \frac{\sum_{j=5}^J \frac{q_{i,t+j}}{(1+r_i)^j}}{GDP_{i,t}} \times 100 \quad (1)$$

where $q_{i,t+j}$ is the annual gross revenue in year $t + j$ from the field discovered in country i in period t , r_i is the annual discount rate for country i , and $GDP_{i,t}$ is annual GDP of country i at year t . The authors use country-specific risk-adjusted discount rates r_i , which are calculated using the relationship between the average of sovereign spreads over a long period, available for a small set of emerging countries, and an index of political risk ratings, available for a wider set of countries. This implies that the $NPV_{i,t}$ measure discounts flows more for countries where political risk is high on average. The sum in the numerator starts at 5 because there is a time delay of 5.4 years on average between a discovery and the start of production in the data. The annual gross revenue $q_{i,t+j}$ is derived from an approximated production profile that starts five years after the announcement of the discovery and up to an exhaustion year J , which is greater than 50 years for a typical giant oil field. Appendix B of [Arezki, Ramey, and Sheng \(2017\)](#) presents a detailed explanation of the approximation of the production profile $q_{i,t+j}$ based on data on existing oil fields at different stages of exhaustion and on details on the process of extraction.⁴ What is essential is that the calculation of $q_{i,t+j}$ is scaled by the size of the discovery in terms of its URR.

³For more details on the construction of the NPV see Section IV.B. in [Arezki, Ramey, and Sheng \(2017\)](#).

⁴Another important assumption that the authors make regarding gross revenues $q_{i,t+j}$ is that the price of oil remains the same for the subsequent years after an oil discovery. This assumption is made for convenience because projecting future oil prices is complicated and oil prices are highly persistent in the data, with a behavior similar to that of a random walk.

Figure 1: Distribution of NPV of giant oil discoveries



Percent of GDP, EMBI countries, 1993–2012. The largest discovery in the sample was in Kazakhstan in 2000 with a NPV of 467.

There are 37 emerging economies with sovereign spreads data in JP Morgan’s Emerging Markets Bonds Index (EMBI), which starts in 1993.⁵ Considering these economies and the years 1993–2012, there are 61 giant oil field discoveries in 15 of the 37 countries. The average and median NPV were 18 and 4.5 percent of GDP, respectively. Figure 1 depicts the distribution of the NPV of these discoveries.

Following the empirical strategy described below, I estimate the effect of giant oil discoveries on sovereign spreads, GDP, the current account, net government debt, and private and government consumption. All macroeconomic data are from the World Development Indicators (WDI) database collected by the World Bank except for net government debt, which is from the Fiscal Monitor Database constructed by the IMF.

2.2 Empirical strategy

Giant oil discoveries have two unique features that allow for the use of a quasi-natural experiment approach to identify their effect. First, while policy and oil prices may drive exploration decisions, the actual timing of discoveries is exogenous due to uncertainty around oil and gas exploration. Second, there is a time delay of 5.4 years on average between discovery and production.⁶ This significant delay allows the treatment of giant oil discoveries as news shocks about higher future

⁵The 37 countries are: Argentina, Belize, Brazil, Bulgaria, Chile, China, Colombia, Dominican Republic, Ecuador, Egypt, El Salvador, Gabon, Ghana, Hungary, Indonesia, Iraq, Jamaica, Kazakhstan, Republic of Korea, Lebanon, Malaysia, Mexico, Pakistan, Panama, Peru, Philippines, Poland, Russian Federation, Serbia, South Africa, Sri Lanka, Tunisia, Turkey, Ukraine, Uruguay, Venezuela, and Vietnam.

⁶Arezki, Ramey, and Sheng (2017) mention that experts’ estimates suggest that it takes between four and six years for a giant oil discovery to go from drilling to production. They also made their own calculation and found that the average delay between discovery and production is 5.4 years.

income.

Following [Arezki, Ramey, and Sheng \(2017\)](#), I estimate the effect of giant oil discoveries using a dynamic panel model with a lag of discovery sizes:

$$y_{i,t} = \rho y_{i,t-1} + \sum_{s=0}^{10} \psi_s NPV_{i,t-s} + \alpha_i + \mu_t + \xi' X_{i,t} + \varepsilon_{i,t} \quad (2)$$

where $y_{i,t}$ is the dependent variable; $NPV_{i,t}$ is the discounted net present value of a discovery in country i in year t ; α_i controls for country fixed effects; μ_t are year fixed effects; $X_{i,t}$ is a vector of additional control variables; and $\varepsilon_{i,t}$ is the error term.⁷

The vector $X_{i,t}$ includes contemporaneous and up to ten lags of the interaction $\mathbb{I}_{disc,i,t-s} * \log p_{oil,t}$, where $\log p_{oil,t}$ is the natural logarithm of the international price of oil in year t and $\mathbb{I}_{disc,i,t-s}$ is an indicator function of whether country i had an oil discovery in period $t - s$. These interaction terms allow the response of $y_{i,t}$ to the price of oil to vary after a discovery in anticipation of higher oil income in the near future.⁸ In addition, the vector $X_{i,t}$ includes the term $R_{i,1993}^{oil} * \Delta \log p_{oil,t}$, where $\Delta \log p_{oil,t}$ is change in the natural logarithm of the international price of oil from period $t - 1$ to t and $R_{i,1993}^{oil}$ are oil rents as a percentage of GDP for country i at the beginning of the sample. This controls for shocks to the price of oil scaled by the sector's relative importance in each economy in 1993, predating all discoveries in the sample of spreads.

2.3 The puzzling response of spreads and borrowing

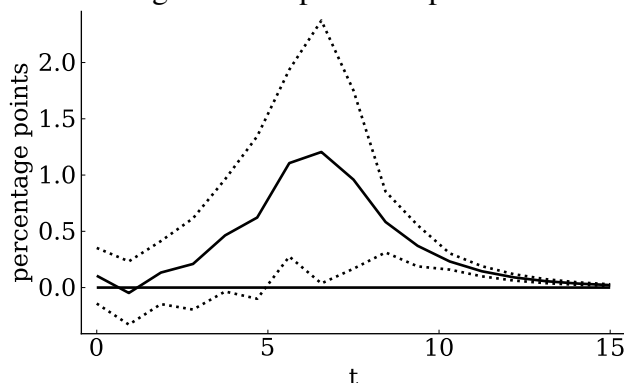
Response in the data.—Figure 2 shows the impulse-response function of spreads to an oil discovery of median size (4.5 percent of GDP).⁹ Following a discovery, spreads steadily increase and, by the sixth year after the discovery was announced, the increase peaks at 1.2 percentage points.

⁷Following [Arezki, Ramey, and Sheng \(2017\)](#), I include country-specific quadratic trends for the regressions of variables $y_{i,t}$ that are non-stationary in the sample. These are GDP, private and government consumption, and spreads. For the first three variables the Im-Pesaran-Shin test ([Im, Pesaran, and Shin \(2003\)](#)) fails to reject the null hypothesis that all panels contain a unit root. The test rejects the null for spreads, but the augmented Dickey-Fuller test ([Dickey and Fuller \(1979\)](#)) on each individual time series of spreads rejects a unit root on only 8 of the 37 series, so out of an abundance of caution I also control for country-specific quadratic trends in this regression.

⁸Appendix A.7 reports every outcome with and without these interactions.

⁹Impulse-response functions are computed as $\Delta y_{i,t} = \rho \Delta y_{i,t-1} + \sum_{s=0}^{10} \psi_s NPV_{i,t-s}$ using the estimated coefficients of equation (2).

Figure 2: Response of spreads

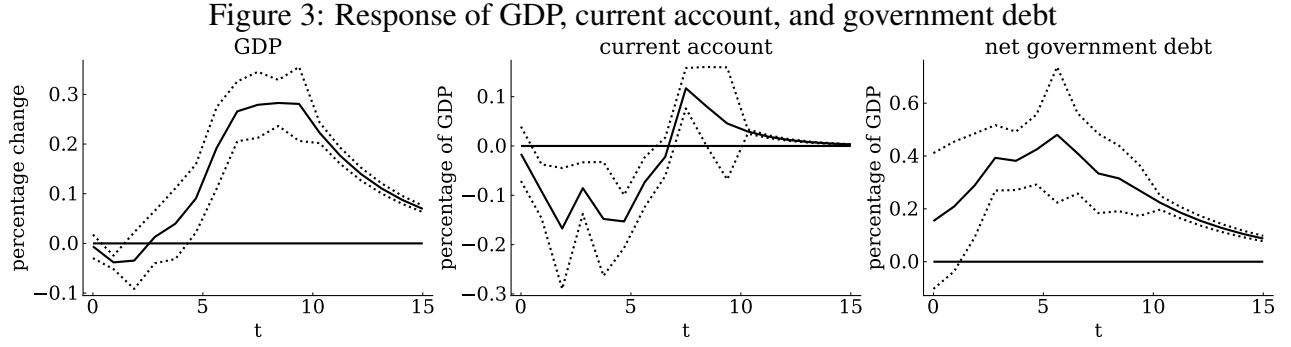


Impulse response to an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

This is the main empirical finding of the paper, and Appendix [A](#) presents multiple robustness exercises. Appendix [A.1](#) shows that this is robust to controlling for proved oil reserves, which have a significant impact on the evolution of credit worthiness of emerging economies who are oil exporters (see [Hamann, Mendez-Vizcaino, Mendoza, and Restrepo-Echavarria \(2023\)](#)). Appendix [A.2](#) presents placebo regressions that show that this response of spreads is not reproduced when discoveries are reassigned at random across countries. This suggests that this estimated response is not driven by unobserved policy changes or global oil-market dynamics that may jointly affect spreads in all countries and the frequency of discoveries (like fiscal expansions from increased exploration incentives). Appendix [A.3](#) shows that there is no evidence of pre-trends in spreads before a discovery. Finally, Appendix [A.4](#) shows that the response is similar under alternative measures of the size of a discovery that do not rely on country-specific discount rates.¹⁰

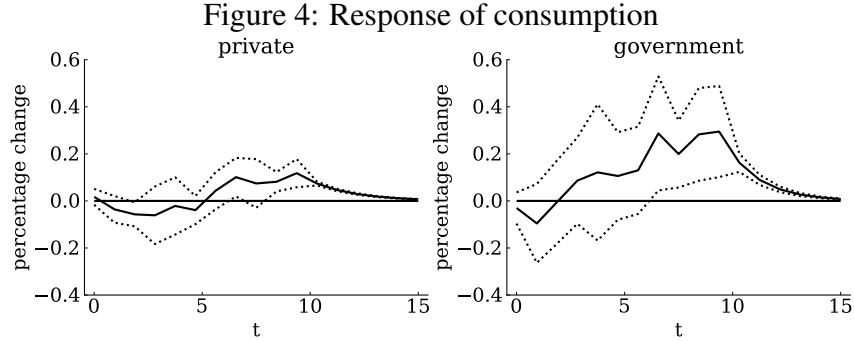
As documented by [Arezki, Ramey, and Sheng \(2017\)](#), the behavior of other macroeconomic aggregates in the data following an oil discovery is consistent with the permanent income hypothesis in an open economy: the economy runs a current account deficit to increase consumption given the higher present value of income, which reverts once the higher income is realized.

¹⁰Appendix [A.6](#) shows that this finding is also robust to dropping the discoveries that occur after the rapid increase in Chinese official lending in 2008 (see Figure 1 in [Horn, Reinhart, and Trebesch \(2021\)](#)). Subsection [4.4](#) further discusses the potential role of official lending.



Impulse response to an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

Figure 3 replicates this part of their analysis and also shows the estimated response of government debt, which increases despite the higher spreads.¹¹ In fact, oil discoveries mostly fuel an increase in government consumption, while the effect on private consumption is modest, as shown in Figure 4.¹²



Impulse response to an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

Response in the canonical model.—Quantitative models of sovereign default perform very well in reproducing the dynamics of borrowing and default risk in emerging economies. As noted by [Arellano \(2008\)](#), and the literature thereafter, a crucial assumption is that the economy suffers a real default penalty that is asymmetric and increasing in output. This assumption, along with

¹¹These estimates use all the available years and countries considered in [Arezki, Ramey, and Sheng \(2017\)](#). Restricting the sample to only the 37 emerging economies in the EMBI yields qualitatively similar, albeit noisier, estimates. These are reported and discussed in Appendix A.5.

¹²[Arezki, Ramey, and Sheng \(2017\)](#) also document an increase in aggregate investment. [Toews and Vézina \(2022\)](#) and [Sheng and Zhao \(2024\)](#) further investigate these effects on investment and show that the increase is mostly driven by Foreign Direct Investment flows. As discussed below, what is key to understanding the response of spreads is the implications of oil discoveries on the cost of default in terms of consumption, regardless of who operates the oil firms. Thus, I abstract from analyzing any form of capital accumulation for simplicity.

persistent output shocks, generates cycles of low spreads and high borrowing when income is high, and capital outflows accompanied by high spreads or default crises when income is low. The response of spreads documented above is puzzling from the point of view of these models because a giant oil discovery is a large positive shock to future income that would reduce default incentives in such a model.

To illustrate this, consider the model in [Arellano \(2008\)](#) augmented with oil income and discoveries. Since I use this model for illustrative purposes, I present here a minimal description of the environment and relegate additional details to Appendix [B](#). There is a small-open economy populated by a household with preferences for consumption represented by $\mathbb{E}_0 [\sum_{t=0}^{\infty} \beta^t u(c_t)]$ and a benevolent government that makes borrowing and default decisions. Each period, the economy receives an income endowment z_t and oil rents R_t^{oil} . Oil rents can take two values

$$R_t^{\text{oil}} = R^{\text{oil}}(n_t) = \begin{cases} R_L^{\text{oil}} & \text{if } n_t \leq T_{\text{wait}} \\ R_H^{\text{oil}} & \text{if } n_t = T_{\text{wait}} + 1, \end{cases} \quad (3)$$

where $R_L^{\text{oil}} < R_H^{\text{oil}}$ and $n_t \in \{-1, 0, 1, \dots, T_{\text{wait}}, T_{\text{wait}} + 1\}$ indicates how many periods have passed since a giant oil field was discovered. The variable n_t follows a Markov chain with transition probability matrix

$$P = \begin{bmatrix} 1 - \pi_{\text{disc}} & \pi_{\text{disc}} & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ \pi_{\text{ex}} & 0 & 0 & \cdots & 0 & 1 - \pi_{\text{ex}} \end{bmatrix}, \quad (4)$$

where π_{disc} is the probability of discovery of a giant oil field and π_{ex} is the probability of exhaustion. This formulation captures the delay between discovery and production that is observed in the data. The government issues one-period non-contingent debt and cannot commit to repay it. Debt is purchased by risk-neutral international investors who behave competitively and consider the government's default incentives when pricing the debt. When the government is in good financial

standing the resource constraint of the economy is

$$c_t + b_t = y_t + q_t b_{t+1}, \quad (5)$$

where q_t is the market price of government debt and $y_t = z_t + R_t^{\text{oil}}$ is total income. If the government defaults then it is excluded from financial markets for a stochastic number of periods and consumption equals total income in default

$$c_t = y_t^{\text{def}}, \quad (6)$$

where $y_t^{\text{def}} = h_z(z_t) + h_{\text{oil}}(R_t^{\text{oil}})$. As discussed in [Arellano \(2008\)](#), the following functional form for h_z generates an asymmetric income cost of default that allows the model to generate procyclical borrowing and countercyclical spreads:

$$h_z(z_t) = \begin{cases} z_t & \text{if } z_t \leq \mu_z \delta_z \\ \mu_z \delta_z & \text{if } z_t > \mu_z \delta_z, \end{cases} \quad (7)$$

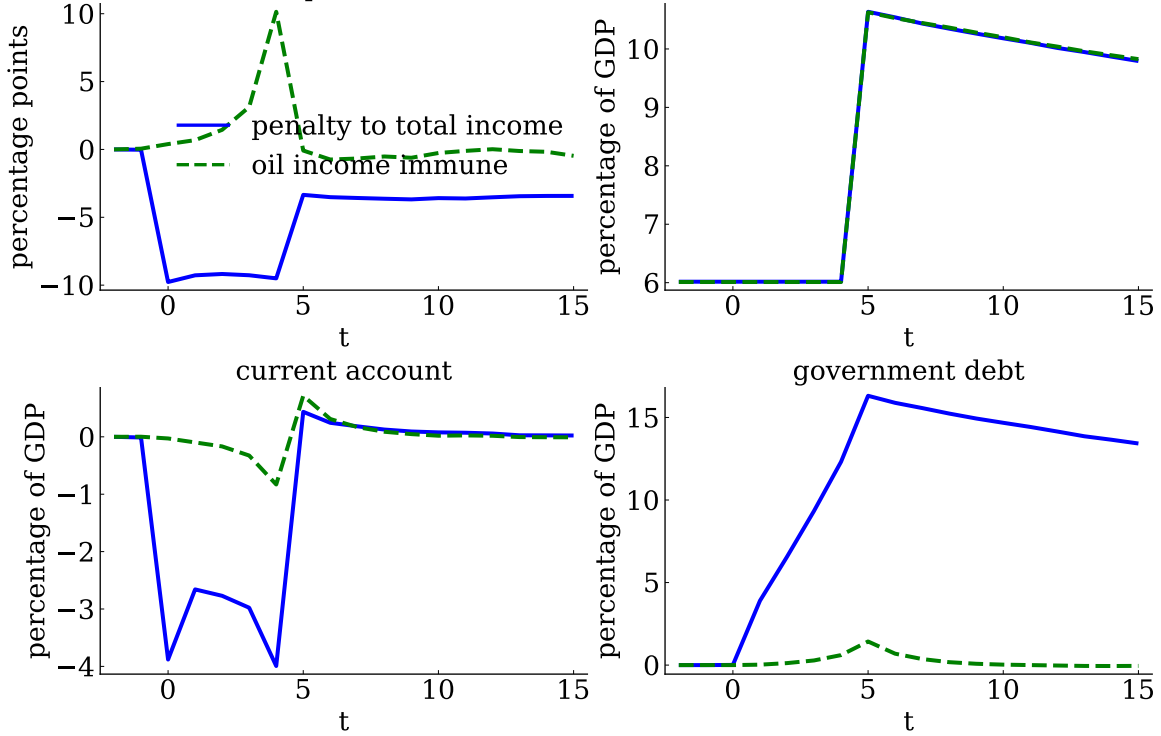
where μ_z is the mean value of z_t and $\delta_z > 0$ controls the severity of the default penalty. For h_{oil} I consider two cases: one in which income from oil rents are equally penalized in default

$$h_{\text{oil}}^{\text{Bench}}(R_t^{\text{oil}}) = \begin{cases} R_t^{\text{oil}} & \text{if } R_t^{\text{oil}} = R_L^{\text{oil}} \delta_{\text{oil}} \\ R_L^{\text{oil}} \delta_{\text{oil}} & \text{if } R_t^{\text{oil}} > R_L^{\text{oil}} \delta_{\text{oil}}, \end{cases} \quad (8)$$

which I refer to as the “canonical model”, and one in which they are immune to the default penalty $h_{\text{oil}}^{\text{Immune}}(R_t^{\text{oil}}) = R_t^{\text{oil}}$. I calibrate the model using the parameter values from [Arellano \(2008\)](#) and setting $R_L^{\text{oil}} = 0.06$ and $R_H^{\text{oil}} = 0.08$, so that oil rents are roughly 6 and 8 percent of average GDP in each regime.¹³ Then, I simulate 10,000 oil discoveries in both economies and compute the average paths of spreads, the current account, and government debt around them. Figure 5 presents the changes in these variables relative to the period before a discovery.

¹³The details of this calibration are in Appendix B.

Figure 5: Average responses to oil discoveries
spreads



Each time series considers 10,000 economies in their ergodic state without any oil discoveries since $t = -50$. The lines are the difference between the values in each period and the values in the first period plotted.

The top-right panel presents the evolution of oil rents, which is by construction identical in both cases. Importantly, when the oil discovery happens in period 0 there is certainty that income will be higher in period 5, so the responses of the current account and government debt are in line with the permanent income hypothesis: the government borrows to front-load consumption given that income in the future will be higher.¹⁴ The stark difference is in the response of spreads. Spreads in the canonical model sharply decrease because the potential penalty of defaulting when $R_5^{\text{oil}} = R_H^{\text{oil}}$ is much higher. The government takes advantage of these more favorable borrowing terms and increases its borrowing by a larger amount. Once $R_5^{\text{oil}} = R_H^{\text{oil}}$ then spreads increase slightly due to the risk of the oil field being exhausted, but they remain lower than before the discovery despite the higher borrowing. This is the prediction from the canonical model that is at odds with the data.

The alternative model in which oil income is immune to the default penalty generates a response of spreads that is in line with the data, while also generating an increase in borrowing, al-

¹⁴The magnitudes are different because the responses correspond to two different environments with the exact same parameter values. What is important is the direction of the responses: an increase in the current-account deficit and in the stock of government debt.

though smaller due to the higher cost from the increase in spreads. The following section presents a richer model in which the default penalty emerges endogenously as a result of financial frictions. Differences in technology and financing requirements in the oil sector imply a relatively small effect of default on oil production, which allows the model to reconcile the responses of spreads to oil discoveries with standard business cycle statistics.

3 Model

The model builds on [Mendoza and Yue \(2012\)](#), who develop a general equilibrium model of both sovereign default and business cycles with endogenous production. Some imported inputs require working capital financing and default triggers an asymmetric efficiency loss when these inputs are replaced by imperfect substitutes as the economy gets excluded from credit markets. I incorporate an oil sector that faces the same financial frictions as the rest of the economy, but allow for the parameters that control its financing needs and technology to be different. In the quantitative analysis in Section 4 I discipline these differences using data for Mexico and elucidate their role in reconciling the empirical findings from Section 2 with standard business cycle statistics.

3.1 Environment

There is a small-open economy populated by a representative household, firms, and a benevolent government. There are three production sectors in the domestic economy: a sector of final goods f , a sector of domestic intermediate goods d , and an oil sector o . Oil and the final good are traded in international markets; domestic intermediates are not.

Final goods sector.—Final goods are produced by a competitive firm owned by the household using a Cobb-Douglas technology:

$$y_{ft} = z_t \left(M_f \left(m_{ft}^d, m_{ft}^* \right) \right)^{\alpha_{Mf}} (L_{ft})^{\alpha_{Lf}} K^{\alpha_K}, \quad (9)$$

where $0 < \alpha_{Mf}, \alpha_{Lf}, \alpha_K < 1$, $\alpha_{Mf} + \alpha_{Lf} + \alpha_K = 1$, L_{ft} is labor hired by the firm for a wage w_t ,

$K = 1$ is fixed, and z_t is a productivity shock that follows an AR(1) process

$$\log z_{t+1} = \rho \log z_t + \sigma_z \varepsilon_{t+1} \quad (10)$$

where ρ is a persistence parameter, and ε_t are iid and follow a standard normal distribution.¹⁵ The function M_f is an Armington aggregator of domestic m_{ft}^d and imported m_{ft}^* materials:

$$M_f(m_{ft}^d, m_{ft}^*) = \left[\lambda_f (m_{ft}^d)^{\frac{\mu-1}{\mu}} + (1 - \lambda_f) (m_{ft}^*)^{\frac{\mu-1}{\mu}} \right]^{\frac{\mu}{\mu-1}} \quad (11)$$

where $\mu > 1$ is the Armington elasticity of substitution and λ_f is the weight of domestic materials.¹⁶

Oil sector.—Oil is produced by a competitive firm owned by the government. This assumption is motivated by the fact that I use Mexico as a calibration benchmark, where oil production is mostly carried out by a government-run company (Petrleos Mexicanos, or PEMEX).¹⁷ The mechanism, however, does not depend on who operates the oil firm. The share of oil rents that accrues to the government affects only the magnitude of the response of spreads to discoveries, as I show in Section 4 below. The oil firm uses technology

$$y_{ot} = z_t \left(M_o(m_{ot}^d, m_{ot}^*) \right)^{\alpha_{Mo}} (L_{ot})^{\alpha_{Lo}} N_t^{\alpha_N} \quad (12)$$

where $0 < \alpha_{Mo}, \alpha_{Lo}, \alpha_N < 1$, $\alpha_{Mo} + \alpha_{Lo} + \alpha_N = 1$, L_{ot} is labor hired by the oil firm, and N_t denotes the size of the available oil field. The function M_o is an Armington aggregator of domestic and

¹⁵I abstract from capital accumulation in all sectors, including oil, out of simplicity. As discussed below, what is crucial for the model to explain the responses of spreads to oil discoveries is the oil sector's exposure to financial frictions relative to that of the rest of the economy.

¹⁶Estimates of the Armington elasticity of substitution between domestic and foreign goods range between 2.5 and 5.1 (see [Bajzik, Havranek, Irsova, and Schwarz \(2020\)](#)).

¹⁷Up until 2014 this company was a constitutionally sanctioned monopoly, with an explicit prohibition of private-sector participation in oil exploration or production. In that year, the Mexican Congress reformed the Constitution to allow for private sector participation, but the industry has continued to be mostly dominated by the same government-run firm.

imported materials similar to that of the final good sector:

$$M_o(m_{ot}^d, m_{ot}^*) = \left[\lambda_o (m_{ot}^d)^{\frac{\mu-1}{\mu}} + (1 - \lambda_o) (m_{ot}^*)^{\frac{\mu-1}{\mu}} \right]^{\frac{\mu}{\mu-1}}, \quad (13)$$

but with a potentially different weight of domestic materials λ_o .¹⁸ The size of the oil field takes two values $N_t \in \{N_L, N_H\}$ with $N_L < N_H$. As with the simple model above, there is an underlying variable $n_t \in \{-1, 0, 1, \dots, T_{\text{wait}}, T_{\text{wait}} + 1\}$ that indicates how many periods have passed since a giant oil field was discovered so that $N_t = N_L$ when $n_t \leq T_{\text{wait}}$ and $N_t = N_H$ when $n_t > T_{\text{wait}}$. This underlying indicator variable n_t follows a Markov chain with the transition matrix described in equation (4).

Imported materials and financial frictions.—Imported inputs m_{it}^* for both $i \in \{f, o\}$ are a Dixit-Stiglitz aggregator

$$m_{it}^* = \left(\int_0^1 (m_{jit}^*)^{\frac{v-1}{v}} dj \right)^{\frac{v}{v-1}} \quad (14)$$

that combines a continuum of differentiated imported varieties m_{jit}^* for $j \in [0, 1]$ with an elasticity of substitution $v > 1$. These inputs are purchased in international markets for prices p_j^* , which are time invariant. Following [Mendoza and Yue \(2012\)](#), I assume that a subset Ω_i of imported varieties needs to be paid for in advance using working capital contracted at a rate r_t^* (these credit contracts are paid within the same period). Let $d_t = 1$ denote that the government is in default and $d_t = 0$ that it is in good financial standing. When $d_t = 0$ then $r_t^* = r^*$ equals the international risk-free rate, which is fixed. When the government is in default $d_t = 1$, the entire economy loses access to global financial markets, which implies that working capital is not available for any of the firms $r_t^* = \infty$.

This assumption is a simplified way of incorporating multiple channels through which a sovereign default may disrupt private access to credit, in particular for working capital. [Arteta and Hale \(2008\)](#) show that sovereign debt crises are accompanied by a decline in foreign credit to domestic private firms that persists for more than two years after a restructuring is agreed. Short-term trade

¹⁸I assume that the Armington elasticity μ is the same in both sectors out of simplicity and because estimates that are specific for the oil sector are unavailable. As [Mendoza and Yue \(2012\)](#) discuss, the main mechanism in the model remains unchanged for any reasonable choice of μ , as well as for the Dixit-Stiglitz curvature parameter v described below.

credit, which is what firms use to pay for imported inputs, is itself restricted during renegotiations (see [Kohlscheen and O'Connell \(2007\)](#)). Banking crises are also associated with sovereign defaults (see [Reinhart \(2010\)](#), [Reinhart and Rogoff \(2011\)](#), and [Laeven and Valencia \(2018\)](#)), and the two reinforce each other: when domestic banks hold government debt, a default deteriorates their balance sheet and contracts domestic credit (see [Bocola \(2016\)](#)). Governments may also impose severe capital controls following a default, as Argentina did in 2002.¹⁹ As in [Mendoza and Yue \(2012\)](#), I model these disruptions in their extreme form, with working capital unavailable to every firm upon default. This severity governs the size of the cost of default, which the calibration targets directly, while the response of spreads to a discovery depends on how differently the two sectors are affected. Subsection 4.3 is devoted to this second margin.

Without loss of generality, let $\Omega_i = [0, \theta_i]$ for $\theta_i \in (0, 1)$. This assumption implies that when access to Ω_i is hampered by exclusion from financial markets imported inputs do not vanish entirely, but rather firms are forced to produce using a less efficient mix. I allow for θ_f and θ_o to be different, which implies that the oil and the final good firms may face a different shadow price for their purchases of imported intermediates

$$P_i^*(d_t) = \left[\int_0^{\theta_i} ((1 + r_t^*) p_j^*)^{1-\nu} dj + \int_{\theta_i}^1 (p_j^*)^{1-\nu} dj \right]^{\frac{1}{1-\nu}} \quad (15)$$

where P_i^* is the standard CES price index. Importantly, allowing θ_f and θ_o to differ allows the oil sector to have different working-capital requirements than the rest of the economy.

Domestic intermediates.—Domestic intermediate goods are produced by a competitive firm owned by the households using labor and technology

$$y_{dt} = A (L_{dt})^\eta, \quad (16)$$

where $\eta \in (0, 1)$, and $A > 0$ captures the presence of fixed sectoral productivity and physical capital installed. The firm sells y_{dt} to the firms in the oil and final goods sectors for a price p_{dt} , which is determined in equilibrium.

¹⁹[Wright \(2006\)](#) studies a model in which capital controls are potentially Pareto-improving when there is a risk that the government will default on behalf of all residents.

Profits.—All firms choose production plans each period to maximize their profits:

$$\begin{aligned}\pi_{ft} &= y_{ft} - p_{dt}m_{ft}^d - P_f^*(r_t^*)m_{ft}^* - w_tL_{ft} \\ \pi_{ot} &= p_{ot}y_{ot} - p_{dt}m_{ot}^d - P_o^*(r_t^*)m_{ot}^* - w_tL_{ot} \\ \pi_{dt} &= p_{dt}y_{dt} - w_tL_{dt},\end{aligned}$$

where π_{ft} , π_{ot} , and π_{dt} are the profits of the final good firm, the oil firm, and the domestic intermediates firm, respectively. The price of the final good is the numeraire and I assume that the price of oil is fixed $p_{ot} = 1$ for simplicity.²⁰

Household.—The household has preferences represented by $\mathbb{E}_0[\sum_{t=0}^{\infty} \beta^t u(c_t, g_t, L_t)]$, where $\beta \in (0, 1)$ is the discount factor, c_t is private consumption of the final good, g_t is government consumption of the final good, and L_t is labor supplied to the domestic firms. The period utility is

$$u(c, g, L) = \frac{\left(c - \frac{L^{1+\frac{1}{\omega}}}{1+\frac{1}{\omega}}\right)^{1-\sigma} - 1}{1-\sigma} + \psi_g \frac{g^{1-\sigma_g} - 1}{1-\sigma_g}, \quad (17)$$

where σ and σ_g are the CRRA parameters for private and government consumption, respectively, ω is the Frisch elasticity of labor supply, and ψ_g is a utility weight on government consumption. I follow [Greenwood, Hercowitz, and Huffman \(1988\)](#) in removing the wealth effect on labor supply by specifying a utility that depends on private consumption net of labor disutility, which is a standard assumption in open-economy business cycle models. The budget constraint of the household is

$$c_t \leq (1 - \tau) [w_tL_t + \pi_{ft} + \pi_{dt}], \quad (18)$$

where τ is an income tax collected by the government.

²⁰Modeling the behavior of the price of oil is a difficult task because of the inherent characteristics of the world market for oil (see [Bornstein, Krusell, and Rebelo \(2022\)](#)). Equation (12) assumes that the oil sector is exposed to the same productivity shock z_t as the rest of the economy. Incorporating exogenous shocks to the price of oil (consistent with the small-open economy assumption) would imply that oil income depends on the product $\tilde{z}_{ot} = z_t p_{ot}$, which is isomorphic to assuming a separate productivity shock to the sector. [Drechsel and Tenreiro \(2018\)](#) develop a small-open economy model in which commodity prices and spreads co-move in a similar fashion as productivity shocks and spreads do. They find that shocks to commodity prices account for a significant share of business cycle fluctuations in emerging economies. Since the purpose of this paper is not to disentangle the role of different shocks (like productivity and oil prices) on driving business cycles, I abstract from these details and let all high-frequency fluctuations (i.e. period by period in the model) to be driven by the single productivity shock z_t .

Government debt and default.—For simplicity, I assume τ is fixed, which is consistent with low-frequency changes in income tax rates. The government issues long-term, non-contingent bonds b_t in international financial markets. Following [Hatchondo and Martinez \(2009\)](#) and [Chatterjee and Eyigungor \(2012\)](#) I assume that bonds mature probabilistically at a rate γ . The law of motion of bonds is:

$$b_{t+1} = (1 - \gamma)b_t + i_{b,t} \quad (19)$$

where $i_{b,t}$ is the amount of bonds issued in period t . When the government is in good financial standing, its budget constraint is

$$g_t + \gamma b_t \leq \tau [w_t L_t + \pi_{ft} + \pi_{dt}] + \pi_{ot} + q_t i_{b,t}, \quad (20)$$

where q_t is the price of government debt that is purchased by risk-neutral competitive foreign investors. If the government defaults then the economy is excluded from financial markets and the default is resolved with probability ϕ in each subsequent period. When the default is resolved the government reenters financial markets with a re-structured stock of debt

$$b_{t+1} = \kappa b_t, \quad (21)$$

where $\kappa \in [0, 1]$ is the recovery rate, which implies a constant haircut of $1 - \kappa$ on the face value of debt.²¹ While the default lasts, creditors hold a defaulted claim that pays no coupon but that retains a market value, which is priced in Subsection 3.2 below. Setting $\kappa = 0$ recovers the standard assumption of reentry with no debt. The government's budget constraint in default is

$$g_t \leq \tau [w_t L_t + \pi_{ft} + \pi_{dt}] + \pi_{ot}. \quad (22)$$

3.2 Recursive formulation and equilibrium

The state of the economy is (b, x) , where $x = (z, n, d)$. Before defining the recursive problem of the government, it is worth noting that the government is the only agent that faces a dynamic problem

²¹I abstract from modeling the details of debt restructuring for simplicity. See [Benjamin and Wright \(2009\)](#), [Dvorkin, Sanchez, Sapriz, and Yurdagul \(2021\)](#), and [Almeida, Esquivel, Kehoe, and Nicolini \(2024\)](#) for more detailed analyses of debt re-negotiations.

(choosing b_{t+1}) so the following definition becomes useful.

Static equilibrium.—Given $x = (z, n, d)$, a *static equilibrium* is allocations for the household $\{c, L\}$ allocations for the final good firm $\{y_f, L_f, m_f^d, m_f^*\}$, allocations for the oil firm $\{y_o, L_o, m_o^d, m_o^*\}$, allocations for the domestic intermediate firm $\{y_d, L_d\}$, and prices $\{p_d, w\}$ such that: (i) given the prices, the allocations for the firms maximize their profits; (ii) the labor supply satisfies the household's first-order condition $L = ((1 - \tau)w)^\omega$; (iii) private consumption satisfies

$$c = (1 - \tau) \left[y_f - m_f^* P_f^*(d) + w L_o + p_d m_o^d \right];$$

and (iv) the prices $\{p_d, w\}$ are such that the markets for labor and the domestic intermediate clear:

$$\begin{aligned} L_f + L_d + L_o &= L \\ m_f^d + m_o^d &= y_d. \end{aligned}$$

As mentioned above, the shadow price of imported intermediates $P_f^*(d)$ depends on the default status. Also, to simplify notation, I omitted the dependence of these equilibrium objects on x , which will be explicit in what follows.

The value of the government at the beginning of a period in good financial standing is

$$V(b, z, n) = \max_{d \in \{0, 1\}} \{dV^D(b, z, n) + (1 - d)V^P(b, z, n)\} \quad (23)$$

where its value in repayment is

$$\begin{aligned} V^P(b, z, n) &= \max_{g, b'} \{u(c(x), g, L(x)) + \beta \mathbb{E}[V(b', z', n')]\} \\ s.t. \quad g + \gamma b &\leq \tau \left[y_f(x) - m_f^*(x) P_f^*(0) + w(x) L_o(x) + p_d(x) m_o^d(x) \right] + \pi_{ot}(x) \\ &\quad + q(b', z, n) [b' - (1 - \gamma)b] \\ \pi_{ot}(x) &= y_o(x) - p_d(x) m_o^d(x) - m_o^*(x) P_o^*(0) - w(x) L_o(x) \\ x &= (z, n, 0) \end{aligned} \quad (24)$$

and its value in default is

$$\begin{aligned}
V^D(b, z, n) &= u(c(x), g, L(x)) + \beta \phi \mathbb{E}[V(\kappa b, z', n')] + \beta (1 - \phi) \mathbb{E}[V^D(b, z', n')] \quad (25) \\
s.t. \quad g &= \tau \left[y_f(x) - m_f^*(x) P_f^*(1) + w(x) L_o(x) + p_d(x) m_o^d(x) \right] + \pi_{ot}(x) \\
\pi_{ot}(x) &= y_o(x) - p_d(x) m_o^d(x) - m_o^*(x) P_o^*(1) - w(x) L_o(x) \\
x &= (z, n, 1).
\end{aligned}$$

Note that the government takes as given the static equilibrium, which only depends on the realized shocks and on the government's default decision.

Lenders payoffs.—The payoff to a lender holding one unit of debt at the beginning of a period in state (b, z, n) is:

$$Q(b, z, n) = (1 - d(b, z, n)) [\gamma + (1 - \gamma) q(b^P(b, z, n), z, n)] + d(b, z, n) q^D(b, z, n), \quad (26)$$

which equals the coupon γ plus the market value of the fraction that does not mature if the government repays ($d(b, z, n) = 0$), and the market value of the defaulted claim if it does not ($d(b, z, n) = 1$). The prices of government debt in repayment and default are, respectively:

$$q(b', z, n) = \frac{\mathbb{E}[Q(b', z', n')]}{1 + r^*} \quad (27)$$

$$q^D(b, z, n) = \frac{\mathbb{E}[\phi \kappa Q(\kappa b, z', n') + (1 - \phi) q^D(b, z', n')]}{1 + r^*}. \quad (28)$$

Equation (28) says that a unit of defaulted debt becomes κ units of a performing bond when the default is resolved, which happens with probability ϕ , and remains a defaulted claim otherwise. Note that the government may default again on the restructured stock κb , in which case the creditors who received the new bonds hold defaulted claims once more. When $\kappa = 0$ equation (28) gives $q^D = 0$ and equation (27) collapses to the usual pricing condition in which defaulted debt is worthless. Now we can define the recursive equilibrium in a standard fashion.

Recursive equilibrium.—A *recursive equilibrium* is value and policy functions V, V^P, V^D, d, b^P, g for the government, and price schedules q and q^D such that: (i) given q and q^D , the value and policy functions solve equations (23), (24), and (25); and (ii) the price schedules satisfy equations

(28) and (27).

3.3 Discussion

The model features two costs from defaulting: temporary financial autarky and inefficient allocations of intermediate imports. The first is exogenous and triggers the second endogenously through the working-capital requirements implied by the financial friction. When access to working capital becomes limited by a default, imported materials in the set Ω_i have to be replaced with imperfect substitutes, which increases the shadow price of the bundle m_i^* . Let the price of all imported varieties j be $p_j^* = 1$. Then, the shadow price that sector i faces for the bundle of imported intermediates is

$$P_i^*(d=0) = \left[\theta_i (1+r^*)^{1-\nu} + (1-\theta_i) \right]^{\frac{1}{1-\nu}},$$

when the government is in good standing, and

$$P_i^*(d=1) = (1-\theta_i)^{\frac{1}{1-\nu}}$$

when the government is in default. It is clear that $P_i^*(d=0) < P_i^*(d=1)$ as long as $\nu > 1$ (which holds by assumption). All else equal, the oil and final good firms demand fewer imported materials. This is the mechanism developed by [Mendoza and Yue \(2012\)](#) through which output drops endogenously in default.²² The key innovation in the model presented above is that this endogenous default penalty may affect each sector differently due to differences in production technologies and in their exposure to the financial friction. As illustrated in Subsection 2.3, whether spreads increase or decrease following an oil discovery depends crucially on how much more or less exposed the oil sector is to the cost of default. The quantitative model builds on a previously studied theory for the cost of default and serves as a structural laboratory to discipline this differentiated sectoral exposure to it.

Given a state in period t , the log-change in output in each sector upon default can be approxi-

²²They also show that this drop is larger for high realizations of z_t due to “strong” convexity of the Cobb-Douglas production function. This asymmetry is key to generating counter-cyclical default risk, as pointed out by [Arellano \(2008\)](#) and the literature thereafter.

mated to a first order as:

$$\Delta \log y_{ft} \approx -\frac{\alpha_{Mf}}{\alpha_K} \times s_{ft}^* \times \left[\frac{-\log(1 - \theta_f)}{\nu - 1} \right] \quad (29)$$

$$\Delta \log y_{ot} \approx -\frac{\alpha_{Mo}}{\alpha_N} \times s_{ot}^* \times \left[\frac{-\log(1 - \theta_o)}{\nu - 1} \right], \quad (30)$$

where

$$s_{ft}^* = \frac{P_{ft}^* m_{ft}^*}{P_{ft}^M M_{ft}} = (1 - \lambda_f)^\mu \left(\frac{P_{ft}^M}{P_{ft}^*} \right)^{\mu-1} \quad (31)$$

$$s_{ot}^* = \frac{P_{ot}^* m_{ot}^*}{P_{ot}^M M_{ot}} = (1 - \lambda_o)^\mu \left(\frac{P_{ot}^M}{P_{ot}^*} \right)^{\mu-1} \quad (32)$$

are the expenditure shares on imported intermediates relative to total expenditure on intermediates in period t .²³ Thus, how differently each sector is affected by default depends, to a first order, on three pairs of parameters: θ_i , which control sector i 's working-capital requirements; $1 - \lambda_i$, which control the relative importance of imported intermediate goods; and α_{Mi} , which control the relative importance of the aggregate intermediate bundle M_i . Based on the above equations, I define

$$\Gamma = \left(\frac{\alpha_{Mo}/\alpha_N}{\alpha_{Mf}/\alpha_K} \right) \times \left(\frac{\bar{s}_o^*}{\bar{s}_f^*} \right) \times \left(\frac{\log(1 - \theta_o)}{\log(1 - \theta_f)} \right) \quad (33)$$

as a sufficient statistic for how the oil sector is affected by default relative to the rest of the economy, where $\bar{s}_o^*$ and $\bar{s}_f^*$ denote average values of the shares in equations (31) and (32).²⁴ This ratio Γ is a state-invariant approximation to $\Delta \log y_{ot}/\Delta \log y_{ft}$, so parameterizations for which $\Gamma < 1$ have the feature that oil is endogenously less affected by default.

In the following section, I discipline these parameters using data from Mexico's input-output table and the financial statements of the government-run oil firm. The Mexican data are a good starting point to illustrate the model's main mechanism because they feature $\theta_o < \theta_f$, $1 - \lambda_o < 1 - \lambda_f$, and $\alpha_{Mo} < \alpha_{Mf}$, which imply $\Gamma < 1$ and allow the model to explain the data as the simple

²³These first-order approximations assume domestic prices, w_t and p_{dt} , remain fixed between default and repayment. This is for convenience in the exposition of the main parameters driving the results. In the following Section I compute these changes exactly considering all the general equilibrium effects.

²⁴This is the ratio of the right-hand side of equation (30) over the right-hand side of equation (29). Subsection 4.3 compares this sufficient statistic to the exact ratio $\Delta \log y_{ot}/\Delta \log y_{ft}$ computed in general equilibrium.

model in Subsection 2.3 suggests. I then compute Γ using data from a cross-section of countries and use it to argue that this mechanism is robust.

4 Quantitative analysis

I solve numerically for the equilibrium using value function iteration. Following Hatchondo, Martinez, and Sapriz (2010), I compute the limit of the finite-horizon version of the economy using a non-linear optimization routine to solve for the borrowing decision in every state. I approximate value functions and the price schedules for debt using linear interpolation, and compute expectations over the productivity shock using a Gauss-Legendre quadrature.

4.1 Calibration

I calibrate the model to the Mexican economy. There are two reasons why Mexico is an ideal example. First, Mexico has been widely studied in the sovereign debt literature because its business cycle has the same properties as other emerging economies (see for example Aguiar and Gopinath (2007), and Aguiar, Chatterjee, Cole, and Stangebye (2016)). As noted by Bianchi, Hatchondo, and Martinez (2018), Mexico gives calibration targets for average levels of debt and spreads that are close to the median value for emerging economies. In short, Mexico is a typical emerging economy so the calibration appeals to the average results from Section 2. Second, Mexico had a series of giant oil discoveries during the 1970s, the largest in 1977 and 1979 with roughly 21 and 12 billion barrels of URR, which imply NPV values of roughly 50 percent of Mexico's GDP each. Five years later, in 1982, Mexico defaulted on its debt. This is an example of the main dynamics studied in this paper.

A period in the model is one year and I mostly follow the calibration strategy in Mendoza and Yue (2012). There are two sets of parameters: the first (summarized in Table 1) is calibrated directly and the second (summarized in Table 2) is chosen using the simulated method of moments (SMM) to target a set of moments from the data. I set the CRRA parameters to $\sigma = \sigma_G = 2$, and the risk free interest rate to $r^* = 0.04$, which are standard values in the sovereign default literature. The Frisch wage elasticity is $\omega = 2.2$, which is a value typically used in RBC models of small open economies (see Mendoza (1991), Neumeier and Perri (2005), and Mendoza and Yue

(2012)). The probability of reentry to financial markets is $\phi = 0.25$ for an average exclusion of 4 years, which is the median value reported in [Gelos, Sahay, and Sandleirs \(2011\)](#) for a large set of default episodes. For the recovery rate of debt κ I build on the work by [Asonuma, Niepelt, and Rancière \(2023\)](#), who compute recovery rates for 531 individual debt instruments in 44 sovereign restructuring episodes between 1999 and 2020. Recovery in their data averages 55 percent across instruments and 66 percent across episodes, so I set $\kappa = 0.60$. I set the income tax rate to $\tau = 0.159$ which is the average of the tax-revenue-to-GDP ratio for Mexico, I normalize $\psi_g = 1.0$, and set the debt-maturity parameter to $\gamma = 0.05$ which gives an average debt maturity of 20 years.

Table 1: Parameters calibrated directly

Parameter		Value	Source
CRRA parameters	$\sigma = \sigma_g$	2.0	Standard value
Frisch wage elasticity	ω	2.2	Standard value
Risk-free rate	r^*	0.04	Standard value
Probability of reentry	ϕ	0.25	4-year average exclusion
Recovery rate in restructuring	κ	0.60	Asonuma, Niepelt, and Rancière (2023)
Income tax rate	τ	0.159	Average tax revenue / GDP
Utility from government consumption	ψ_g	1.0	Normalization
Maturity of debt	γ	0.05	Mexico's Ministry of Finance
Share of intermediates in final sector	α_{Mf}	0.46	Mexico's I-O table
Labor share in final sector	α_{Lf}	0.14	Mexico's I-O table
Capital share in final sector	α_K	0.40	Mexico's I-O table
Share of intermediates in oil sector	α_{Mo}	0.30	Mexico's I-O table
Labor share in oil sector	α_{Lo}	0.04	Mexico's I-O table
Resource share in oil sector	α_N	0.66	Mexico's I-O table
Armington elasticity	μ	2.9	Mendoza and Yue (2012)
Weight of domestic inputs, final sector	λ_f	0.62	Mendoza and Yue (2012)
Weight of domestic inputs, oil sector	λ_o	0.656	Mexico's I-O table
Dixit-Stiglitz curvature parameter	ν	2.44	Mendoza and Yue (2012)
Labor share for domestic intermediates	η	0.26	$\frac{\alpha_{Lf}}{\alpha_{Lf} + \alpha_K}$
Size of small oil field	N_L	0.025	Oil sector 6% of GDP
Size of giant oil field	N_H	0.045	69% increase in oil share of GDP
Probability of discovery	π_{disc}	0.01	Oil discoveries data
Delay between discovery and production	T_{wait}	5	Oil discoveries data
Probability of exhaustion	π_{ex}	0.02	Oil discoveries data
Persistence of z_t	ρ	0.95	Standard value
Working capital requirement for oil sector	θ_o	0.36	PEMEX financial statements

The size of the small oil field is set to $N_L = 0.025$ so that steady-state GDP in the oil sector represents 6 percent of total GDP, which is the average long-run value for Mexico since 1993. The average giant oil discovery in the data is 3.8 billion barrels of ultimately recoverable reserves

and Mexico's proven reserves are 5.5 billion barrels, so an average discovery would imply a 69 percent increase in Mexico's current oil production capacity.²⁵ Thus, I set $N_H = 0.045$ so that the oil sector's share of steady-state GDP is 69 percent higher than with $N_t = N_L$, rising from 6.00 to 10.14 percent.²⁶ The probability of a discovery is $\pi_{\text{disc}} = 0.01$, which is the probability of new discoveries in the data. The probability of exhaustion is $\pi_{\text{ex}} = 0.02$ for an average field life of 50 years. I set the persistence of the productivity shock to $\rho = 0.95$, which is a standard value (the volatility is computed as part of the SMM exercise described below).

To calibrate the technology parameters I rely on Mexico's Input-Output matrix for 2018, which is disaggregated at the Subsector level using the North American Industry Classification System (NAICS). I separately consider Subsector 211, which corresponds to Oil and Gas Extraction, and the total economy minus this Subsector 211. The shares of intermediate goods in gross output are $\alpha_{Mf} = 0.46$ and $\alpha_{Mo} = 0.30$. I set $\alpha_{Lf} = 0.14$ and $\alpha_{Lo} = 0.04$ as the ratios of labor compensation to the value of production, and $\alpha_K = 1 - \alpha_{Mf} - \alpha_{Lf}$ and $\alpha_N = 1 - \alpha_{Mo} - \alpha_{Lo}$. The value for α_{Lf} implies a labor share on value added of 0.26, which is low relative to what it is in advanced economies but in line with other estimates of the labor share in emerging economies (for example, [Karabarbounis and Neiman \(2013\)](#) estimate that the total labor share of Mexico declined from 0.38 to 0.28 between 1993 and 2011). I set the labor share in the domestic intermediates sector to be the same as in the final goods sector $\eta = 0.26$. I take the Dixit-Stiglitz curvature parameter $\nu = 2.44$ from [Mendoza and Yue \(2012\)](#), as well as the values for the Armington elasticity of substitution $\mu = 2.9$ and the weight $\lambda_f = 0.62$, which they estimate jointly from time-series data.²⁷ To calibrate λ_o (importantly, to calibrate how different it is from λ_f) I use the following ratio from firms' first-order conditions

$$\frac{s_{ft}^d / (1 - s_{ft}^d)}{s_{ot}^d / (1 - s_{ot}^d)} = \left(\frac{\lambda_f}{1 - \lambda_f} \frac{1 - \lambda_o}{\lambda_o} \right)^\mu \left(\frac{P_{ft}^*}{P_{ot}^*} \right)^{\mu-1}, \quad (34)$$

where s_{ft}^d and s_{ot}^d are the shares of each sector's purchases of intermediates that are domestic. These shares in the Mexican I-O matrix are 0.617 and 0.718, respectively. For the purposes of the calibration exercise, I set the relative price P_{ft}^*/P_{ot}^* equal to one, which implies a value of

²⁵This estimate is from the OPEC annual statistical bulletin, Table 3.1.

²⁶Oil output itself rises by 77 percent, because total GDP also grows when the giant field produces.

²⁷Appendix C shows that the main mechanism discussed in Subsection 4.3 is not sensitive to either elasticity of substitution.

$$\lambda_o = 0.656.^{28}$$

The financial friction is controlled by the shares of imported intermediates that have to be purchased with working capital θ_f and θ_o . With respect to θ_o , the fact that oil production in Mexico is mostly dominated by a single government-run firm presents an advantage because the firm's financial statements are publicly available. I use PEMEX's audited financial statements for 2018 (the same year as the input-output matrix) and compute the ratio of debt to suppliers (reported as a part of short-term liabilities in the balance sheet) divided by the total cost of sales, which is equal to 0.125.²⁹ The value for $\lambda_o = 0.656$ above implies that a fraction 0.344 of materials are imported, so assuming that only imported materials require working capital (as in the model) I get a value for $\theta_o = 0.125/0.344 = 0.36$. Note that this assumption is rather strong, but conservative from the point of view of the mechanisms in place (a lower value of θ_o would imply that the oil sector is even less affected by default). Calibrating θ_f presents challenges due to data availability. Detailed data on working capital at the aggregate level is scarce for emerging economies, which is why [Mendoza and Yue \(2012\)](#) set θ_f as part of the SMM exercise. I do the same.

Table 2: Parameters set with SMM

Parameter	Value	Targets			
		Moment	Data	Model	
Standard deviation of z_t	σ_z	0.021	GDP standard deviation	3.11	3.06
TFP of domestic intermediates	A	0.069	Output drop in default	-13.28	-14.47
Discount factor	β	0.922	Average spread	2.90	2.24
Working capital requirement for final sector	θ_f	0.552	Private working capital / GDP	8.09	7.94

Moments are expressed in percentage points. Moments for GDP, spreads, and working capital are computed by simulating 2,000 time series of 400 periods after trimming the first 100 without any oil discoveries and that are in good financial standing in the initial period. For the output drop in default, I simulate 2,000 time series of 20,000 periods and compute the output drop whenever a default episode happens. GDP data are applied the HP Filter with a smoothing parameter of 100.

Table 2 summarizes the SMM exercise, which consists in choosing $(\sigma_z, A, \beta, \theta_f)$ jointly to target a standard deviation of GDP of 3.1 percent, an average drop of output in default of 13.3 percent, an average spread of 2.9 percent, and a share of working capital financing in GDP of 8.1 percent. The standard deviation of GDP corresponds to the cyclical component of real GDP in

²⁸Since $P_i^*(d=0) = \left[\theta_i (1+r^*)^{1-\nu} + (1-\theta_i) \right]^{\frac{1}{1-\nu}}$ is monotone in θ_i , this ratio of prices lies between $1/(1+r^*)$ and $1+r^*$ for any working-capital requirements. Setting it to one therefore misstates the right-hand side of equation (34) by at most 2.6 percent. At the calibrated values this error is 0.5 percent, which would change λ_o by 0.001.

²⁹This is credit extended to the firm by its suppliers, not a transfer from the government. Note that the government cannot substitute for this credit during a default episode, because it is itself excluded from credit markets.

local currency units after applying an HP filter with a smoothing parameter of 100 using annual data from 1960 to 2019. As mentioned above, Mexico experienced a sovereign default in August 1982. The log deviation in 1983 from the pre-default linear trend is -0.133 (Arellano (2008) and Mendoza and Yue (2012) report a similar drop for the Argentinean default of 2001). The average spread is from the EMBI data used in Section 2 for the years between 1997 and 2019. Finally, I follow Schmitt-Grohe and Uribe (2007) and proxy working capital in the private sector as the fraction of M1 held by firms with an estimate for the U.S. that shows firms own about two-thirds of M1. Using Mexico's M1 data and the same two-thirds I get a target of 8.1 percent (Mendoza and Yue (2012) get 6 percent for Argentina).

4.2 Model fit

The quantitative model performs well in fitting untargeted business-cycle moments, as previous models in the literature do. In addition, this section shows that the model also fits the responses to oil discoveries documented in Section 2 through the mechanism discussed in Subsection 3.3.

Business-cycle moments.—Table 3 compares business-cycle moments from the data to those from different calibrations of the model. The benchmark corresponds to the calibration to Mexico described above and the remaining rows are alternative technologies for the two sectors that I study in Subsection 4.3. The benchmark model performs well in replicating most of the untargeted Mexican business-cycle co-movements, as well as the average level of external government debt. Spreads in the model are slightly more volatile than in the data, 1.87 against 1.36. The dimensions in which the model falls short are the volatilities of the trade balance and the current account, and the volatility of private consumption relative to output. This is a result of the relatively small size of the government, along with the assumption that there are no lump-sum taxes or transfers. The model also abstracts from other important sources of external imbalances like private borrowing.

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³⁰Mendoza and Yue (2012) circumvent this problem by assuming an exogenous quantity of resources being traded each period. They also show that this is inconsequential to the behavior of most of the other variables, so I choose to omit them for simplicity. Appendix D shows that the main results do not change with a version of the model that consolidates the resource constraint, and that this version accounts for a large part of the gap in these volatilities.

Table 3: Business cycle moments

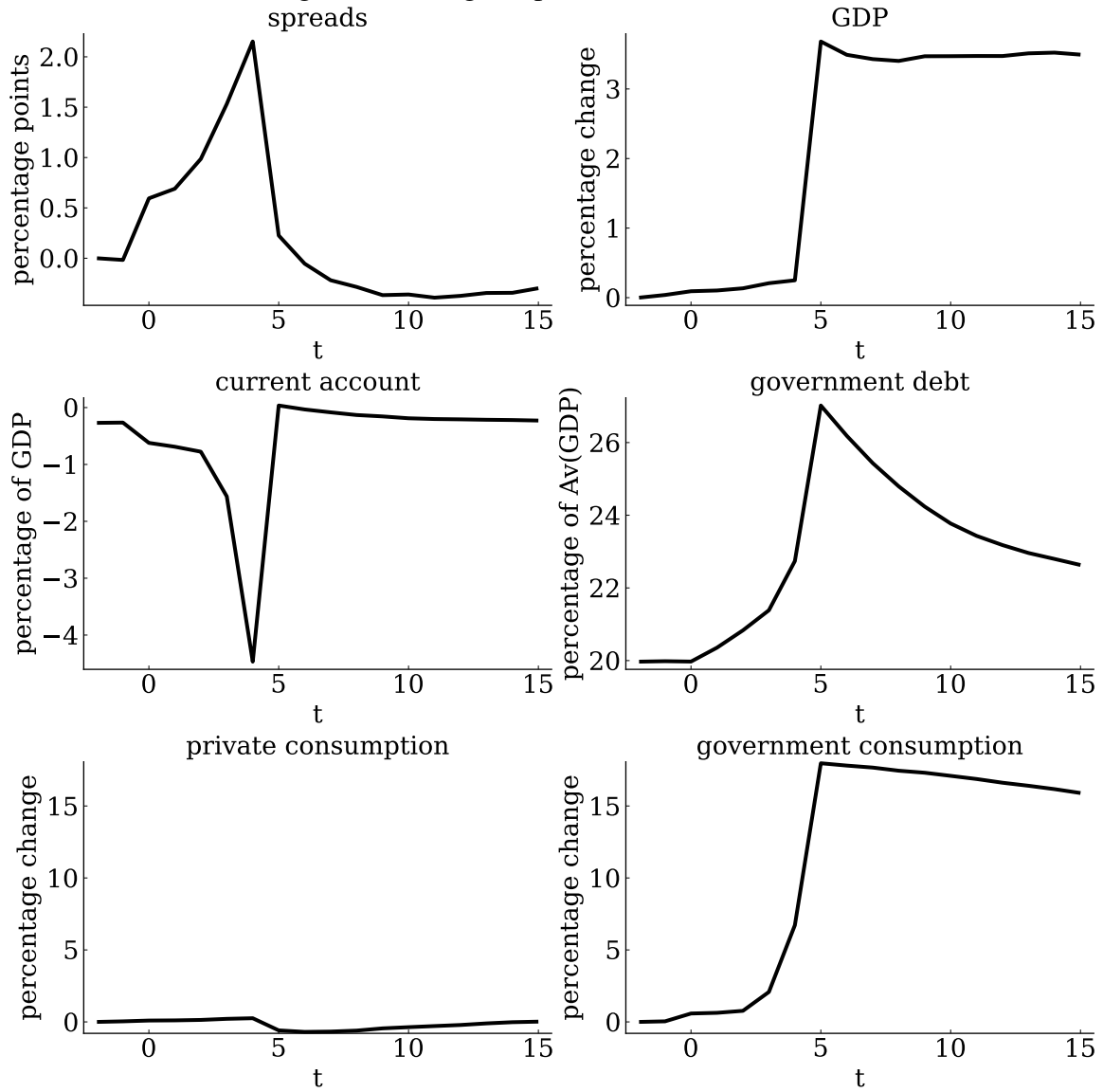
	σ_y	Δy^{def}	μ_{r-r^*}	$\frac{wk}{y}$	$Pr(d=1)$	$\frac{b}{y}$	σ_{r-r^*}	$\frac{\sigma_c}{\sigma_y}$	$\frac{\sigma_g}{\sigma_y}$	$\frac{\sigma_{tb}}{y}$	$\frac{\sigma_{ca}}{y}$	$\rho_{r-r^*,y}$	$\rho_{\frac{tb}{y},y}$	$\rho_{\frac{ca}{y},y}$
data	3.11	-13.28	2.90	8.09	2.45	0.21	1.36	1.45	1.05	2.57	2.14	-0.22	-0.19	-0.49
benchmark	3.06	-14.47	2.24	7.94	3.02	0.19	1.87	1.01	1.04	0.28	0.83	-0.25	-0.27	-0.09
$\theta_o = \theta_f$	3.05	-14.65	2.19	7.94	3.05	0.20	1.86	1.01	1.05	0.29	0.87	-0.25	-0.27	-0.09
identical	3.11	-15.47	2.05	7.99	2.71	0.24	1.79	0.98	1.18	0.34	1.00	-0.25	-0.29	-0.10
Brazil	3.21	-17.07	1.81	8.09	2.46	0.36	1.75	0.95	1.35	0.47	1.46	-0.23	-0.29	-0.09

The first four columns are the moments targeted by the calibration of Table 2. Moments are computed by simulating 2,000 time series of 400 periods after trimming the first 100 without any oil discoveries and that are in good financial standing in the initial period. Periods spent in default are excluded when computing the moments. GDP and consumption data are applied the HP Filter with a smoothing parameter of 100. All moments are in percent, except $\frac{b}{y}$ and the ratios of standard deviations and the correlations.

Importantly, the model is successful in generating countercyclical spreads and a countercyclical trade balance (procyclical borrowing). The correlation between the current account and output is negative as well, although weaker than in the data. As it has been discussed in the literature, this feature is a direct result of the asymmetric default penalty on output (see [Arellano \(2008\)](#) and [Mendoza and Yue \(2012\)](#) for discussions of this mechanism, and [Aguiar and Amador \(2014\)](#) for an extended survey). A key observation here is that, as the spreads data for Mexico, these model samples do not include oil discoveries.

Responses to oil discoveries.—Figure 6 shows that spreads in the model increase following an oil discovery, along with borrowing and GDP, which are the puzzling joint responses documented in Section 2. Each time series averages 10,000 paths around an oil discovery in period $t = 0$. The state in the initial period $t = -1$ of each path is a draw from the ergodic distribution with $N_t = N_L$, without any oil discoveries in the past 50 periods.

Figure 6: Average responses to oil discoveries
spreads



Each time series considers 10,000 economies in their ergodic state without any oil discoveries since $t = -50$. The lines are the difference between the values in each period and the values in the first period plotted.

As in the data, spreads start to increase when news of a discovery arrives and peak in $t = 4$ right before production in the new field starts in $t = 5$. The responses of the other variables are in line with those in the data presented in Section 2. Despite higher spreads, government borrowing increases as a response to the large future increase in income from higher oil production, driving a current account deficit. Once the field becomes productive in period $t = 5$ the current account reverts, and both GDP and government consumption increase, while private consumption remains mostly unchanged. The government also uses this higher income to reduce the debt, which drives

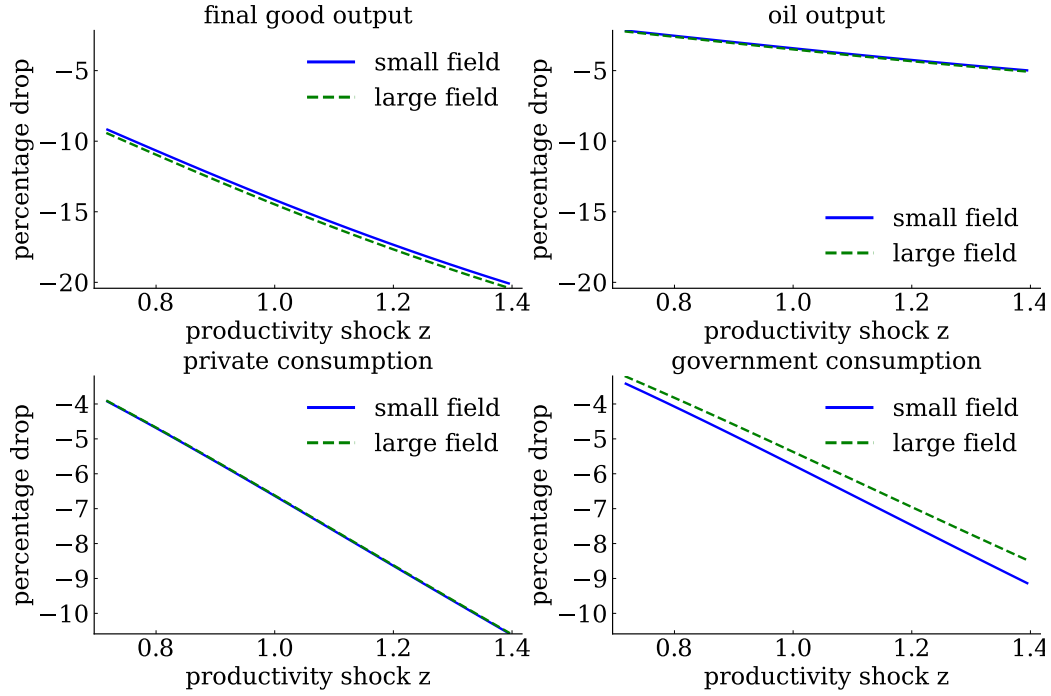
the decrease in spreads following the start of production in the new field.

4.3 The role of technology and working-capital requirements

This subsection studies how the response of spreads to a discovery depends on the technology and working-capital parameters of the two sectors. I first show how these parameters shape the cost of default under the benchmark calibration. I then compute the sufficient statistic Γ for a cross-section of oil-producing countries, show that it is smaller than one for most of them, and show that it is a good proxy for the exact relative exposure in the model. As discussed in Subsection 3.3, $\Gamma < 1$ means that the oil sector is relatively less exposed to the endogenous cost of default, which is the property that drives the result in the simple model of Subsection 2.3. Finally, I solve the model under alternative technologies and show that $\Gamma < 1$ is a sufficient but not a necessary condition for spreads to increase following a discovery.

Figure 7 presents the endogenous cost of default in terms of output (top panels) and in terms of consumption (bottom panels). First, note that as in Mendoza and Yue (2012) the cost is proportionately larger with high realizations of the productivity shock. Second, note that the cost of default in the oil sector is significantly smaller than in the rest of the economy. When the oil firm is operating a larger field then a larger fraction of government income depends on the sector that is less affected by default. This translates into a smaller (government) consumption-cost of default when the oil field is large, as the two bottom panels show.

Figure 7: Cost of default



All else equal, the government has stronger incentives to default when $N_t = N_H$ because the additional income from oil rents allows it to “weather the storm” of lower tax revenue due to lower non-oil output in default. This is a direct result of the oil sector using fewer intermediate goods $\alpha_{Mo} < \alpha_{Mf}$, fewer imported intermediates $1 - \lambda_o < 1 - \lambda_f$, and requiring working capital for a smaller fraction of its imported materials $\theta_o < \theta_f$. As discussed in Subsection 3.3, the way in which these parameters affect the relative exposure of the two sectors can be summarized by the sufficient statistic Γ proposed in equation (33). Importantly, note that this sufficient statistic illustrates that it is not necessary that all three sectoral differences hold for oil to be less exposed to the cost of defaulting, as it happens to be the case for the Mexican data.

Table 4 computes measures of Γ for multiple oil-producing countries using data from the OECD Inter-Country Input-Output tables. Columns (1) and (2) present the values of α_{Mf} and α_{Mo} as the shares of intermediate goods in gross output for the total economy (minus the oil sector) and the oil sector, respectively. Columns (3) and (4) present the corresponding expenditure shares of imported intermediates out of total expenditure on intermediates, s_f^* and s_o^* , and columns (5) and (6) present two values of the sufficient statistic for different assumptions for the financial friction. Column (5) computes Γ assuming that all sectors are equally affected by the financial friction $\theta_o = \theta_f$, while column (6) computes it for the values of θ_o and θ_f that were calibrated for

Mexico.³¹

Table 4: Technology of oil extraction and of the rest of the economy

Country	Year	Intermediates / gross output		Imported share of intermediates		Sufficient statistic Γ	
		α_{Mf} (1)	α_{Mo} (2)	s_f^* (3)	s_o^* (4)	$\theta_o = \theta_f$ (5)	Calibrated θ_i (6)
Russia	2019	0.52	0.10	0.10	0.05	0.04	0.02
Norway	2020	0.50	0.11	0.22	0.15	0.07	0.04
Thailand	2010	0.64	0.26	0.29	0.27	0.12	0.07
Netherlands	2022	0.56	0.23	0.29	0.26	0.15	0.08
Australia	2022	0.51	0.24	0.10	0.09	0.19	0.10
Mexico	2018	0.46	0.30	0.38	0.28	0.29	0.16
Ukraine	2019	0.60	0.37	0.20	0.24	0.34	0.19
Canada	2021	0.47	0.36	0.21	0.19	0.44	0.25
Peru	2007	0.48	0.38	0.16	0.14	0.45	0.25
Kazakhstan	2022	0.39	0.43	0.23	0.17	0.74	0.42
Colombia	2022	0.51	0.49	0.15	0.18	0.88	0.49
Indonesia	2016	0.47	0.53	0.12	0.13	1.08	0.61
Malaysia	2020	0.59	0.76	0.25	0.20	1.36	0.76
United States	2021	0.45	0.52	0.08	0.16	2.25	1.26
Brazil	2015	0.49	0.61	0.10	0.19	2.53	1.42

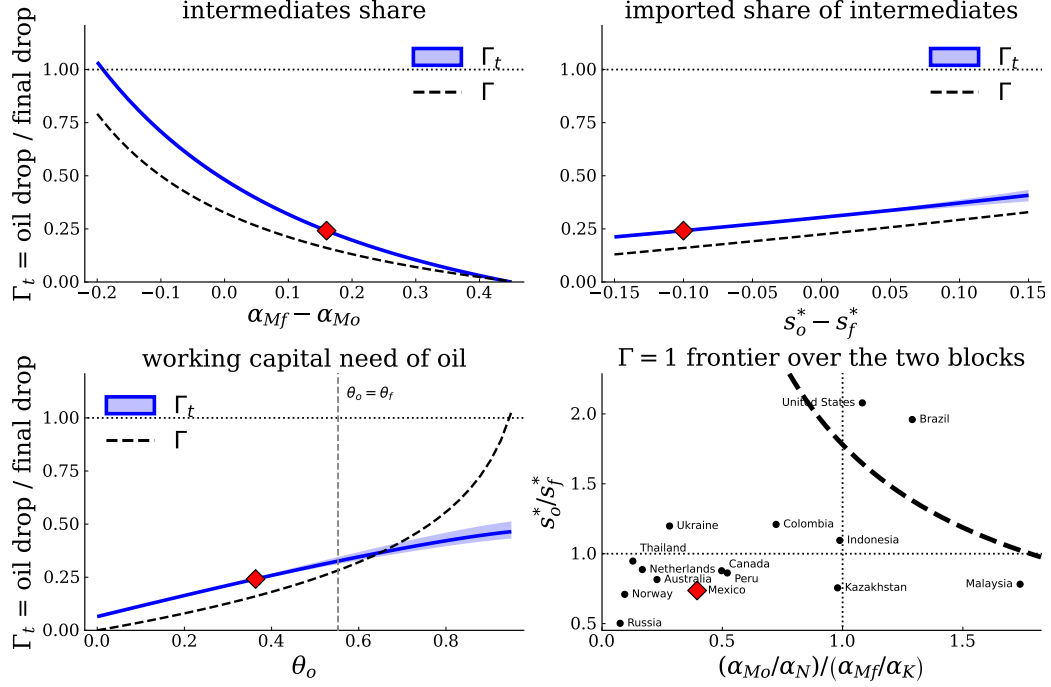
The data for Mexico is taken from INEGI's input-output table for 2018 that was used to calibrate the model. Data for all other countries are from the OECD Inter-Country Input-Output tables. I consider economies with an oil sector of at least one percent of gross output. The year indicates the most recent year for which the OECD reports a measured import use table rather than an imputed one. The oil sector is ISIC Rev.4 Division 06, "Extraction of crude petroleum and natural gas". The last two columns are the sufficient statistic Γ of equation (33) evaluated at each country's own shares and considering two cases for the ratio θ_o/θ_f : setting it equal to 1 (same financial friction in all sectors) and for the values calibrated for Mexico. The labor shares are held at the benchmark calibrated values throughout, so the shares of the fixed factors follow each country's own α_M .

In the model, the oil sector is less exposed to the default penalty for values of $\Gamma < 1$. This is true for 11 of the 15 countries in column (5) of Table 4, which is the conservative case with $\theta_o = \theta_f$. In column (6), which assumes that the degree of financial frictions in Mexico is representative, all but two countries feature a value of $\Gamma < 1$: Brazil and the United States.³²

³¹As mentioned before, these two parameters are difficult to pin-down directly from data. This is why [Mendoza and Yue \(2012\)](#) and I calibrate θ_f as part of the SMM exercise. In the case of Mexico θ_o can be inferred from the financial statements of one firm, but this strategy may not be easily extended to other countries.

³² Γ understates the exact ratio Γ_t along both technology differences, so countries with values close to one may in fact have $\Gamma_t > 1$.

Figure 8: Relative exposure to default and technology gaps



All values are computed from the static equilibrium given the shock and the large oil field. Solid blue lines are Γ_t evaluated at $z_t = 1$ and the shaded areas its range over the grid for z_t . Dashed black lines in the top and bottom-left panels are the sufficient statistic Γ of equation (33). Red diamonds are the benchmark calibration for Mexico. The bottom-right panel presents the two blocks of equation (33) that are measured by input-output tables, with each country at its own values. The dashed black line is the $\Gamma = 1$ frontier, computed at the calibrated θ_i s. Labeled points are the countries in Table 4.

Figure 8 makes three points. First, it shows that Γ is an informative sufficient statistic. The dashed black lines in the top two and the bottom-left panels are values of Γ for different values of the differences in technologies and financial frictions. The solid-blue lines are the values of the exact relative change in output upon default

$$\Gamma_t = \frac{\Delta \log y_{ot}}{\Delta \log y_{ft}}, \quad (35)$$

for a realization of the shock equal to its mean $z_t = 1$. The shaded blue areas are this exact value over the support of z_t considered in the numerical exercises.³³ The values of Γ move with the exact ratios Γ_t for all possible shock realizations and are close to parallel in the top two panels, which indicates that the sufficient statistic is a good proxy for how the actual ratio in general equilibrium

³³I evaluate the shock z in a grid from 0.72 to 1.40, which spans five unconditional standard deviations of $\log z_t$ on each side of its mean. The axes for the two technology differences contain the observed ranges from Table 4. In the bottom-left panel I consider values of θ_o from zero to almost one, since this parameter is only measured for Mexico.

changes with parameters. Along both technology differences $\Gamma < \Gamma_t$, and the two only cross for large values of θ_o . Note, however, that for large values of θ_o the statistic Γ overstates the oil sector's relative exposure, which is much lower in the actual equilibrium.

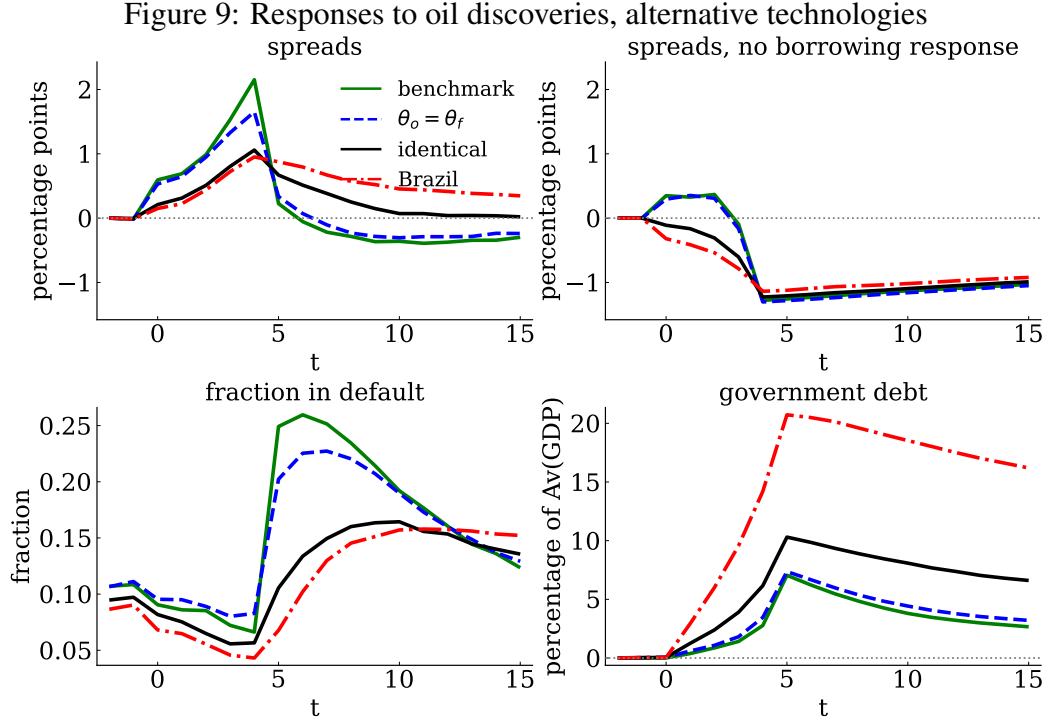
The second point is that Mexico (the red diamond) is not an outlier calibration. This is clearly illustrated by the bottom-right panel, which places the countries in Table 4 on the plane of the two blocks of equation (33) that an input-output table measures directly: the relative intensity of intermediate materials $(\alpha_{Mo}/\alpha_N)/(\alpha_{Mf}/\alpha_K)$ on the horizontal axis and the relative share of imported materials s_o^*/s_f^* on the vertical axis. Keeping the financial frictions fixed at the levels calibrated for Mexico, Γ is the product of these two ratios and a constant of 0.56, so each dot is placed at the value of Γ reported in column (6) of Table 4. The thicker dashed black line in the bottom-right panel is the threshold at which Γ becomes larger than 1 (for values to the north-east of this line). Mexico sits comfortably to the south-west of this threshold, as do most other countries.

Finally, the third point is that the actual ratio Γ_t , which illustrates the main mechanism in the model, is significantly more sensitive to sectoral differences in the share of intermediate materials $\alpha_{Mf} - \alpha_{Mo}$ than to differences in the shares of imported materials or exposure to the financial friction. In fact, neither of these two differences brings Γ_t to one anywhere in the ranges I consider. This is important because, out of the three pairs of parameters, the share of intermediate materials is the parameter for which there are more reliable standardized cross-country national-accounts data.

I now explore how the response of spreads to oil discoveries changes with different values of the parameters that drive Γ_t . I consider three cases: (i) a case for Mexico with no difference in the exposure to the financial friction $\theta_o = \theta_f$, (ii) a case in which both sectors have identical technologies and exposure to the financial friction, and (iii) a case using the technology parameters for Brazil with $\theta_o = \theta_f$. All three cases keep the other parameters set with the simulated method of moments at their benchmark values. The bottom rows in Table 3 show that the moments change little across the first two cases. The one calibrated to Brazil differs more because its final sector buys more intermediates than Mexico's, which makes a default costlier, lowers its spreads and raises the debt-carrying capacity. The debt-to-GDP ratio increases progressively through the cases, as explained below.

Figure 9 presents average responses to a discovery for the benchmark calibration and the three

alternative cases, following the same procedure as the one described for Figure 6. The top-left panel shows the average response of spreads, the top-right panel shows how the spread schedule evolves with N_t , keeping the initial state constant, the bottom-left panel shows the fraction of economies in default, and the bottom-right panel shows government debt as a fraction of GDP.



Each time series considers 10,000 economies in their ergodic state without any oil discoveries since $t = -50$. The lines are the difference between the values in each period and the values in the first period plotted, except in the bottom-left panel, which is a level.

The price schedule when $\theta_o = \theta_f$ (dashed blue line) deteriorates by almost as much as in the benchmark (solid green line). This shows that the sectoral difference in exposure to the financial friction observed for Mexico is not an essential driver of the mechanism.³⁴ The actual response of spreads is slightly smaller, 1.65 against 2.15 percentage points, because the government faces better borrowing terms for the new higher levels of debt. For these two cases, the top two panels show that the increase in spreads is driven both by a shift in the price schedule of debt and by the increase in borrowing shown in the bottom-right panel. The bottom-left panel shows the fraction of economies that are in default following an oil discovery, which increases once the government starts to receive larger oil rents from the field in $t = 5$. Since the debt is long-term, this higher

³⁴This is suggested by columns (5) and (6) of Table 4 that show that Γ only increases from 0.16 to 0.28 for Mexico when $\theta_o = \theta_f$. This is also consistent with the bottom-left panel of Figure 8.

default risk in $t = 5$ is priced-in in all preceding periods starting when the discovery happens in $t = 0$ and spreads gradually increase as higher default risk becomes more imminent.

When both sectors have identical exposure to default (solid black lines) spreads also increase, but the increase is entirely driven by borrowing. When both sectors are equally exposed to the default penalty, the price schedule does not deteriorate at all and in fact improves slightly. The dash-dot red lines correspond to the case calibrated to Brazil. The top-right panel shows that borrowing terms improve following a discovery, as $\Gamma > 1$ suggests, but spreads still increase because that improvement is in tension with a more aggressive increase in borrowing.

These exercises show that $\Gamma < 1$ is a sufficient but not a necessary condition for spreads to increase after a discovery. Identical technologies do not flip the response because the default penalty in this model is a proportional loss of output and not a cap on income. A discovery still raises income in both default and repayment, and the government still borrows against it. In the canonical simple model of Subsection 2.3 spreads fall because the penalty removes the entire increment in oil rents, which implies that a discovery makes the oil sector relatively more exposed to the default penalty. The counterpart of that case here is not $\Gamma = 1$ but a higher threshold, which none of the technologies I consider reaches.

Under the Mexican calibration, the sectoral composition of the default penalty accounts for the entire deterioration in the price schedule that the government faces (top-right panel) and for about half of the increase in observed spreads (top-left panel). With identical technologies the peak response of spreads is 1.06 percentage points, against 2.15 in the benchmark. Across the four economies in Figure 9 both the shift in the schedule and the total response of spreads are monotone in Γ , but the improvement in the schedule is partly offset by an endogenous increase in borrowing. How much the government front-loads is governed by the relatively low discount factor β , which is a simplification of forces outside the model that also shape borrowing against future oil income. I discuss some of these and other mechanisms in the next Subsection. Regardless of which of these forces is at work, the values in Table 4 indicate that the sufficient condition for spreads to increase is satisfied in most oil-producing economies, and that increase would be partly driven by the mechanism presented in this paper.

Finally, note that Mexico's oil monopoly allows for a direct calibration of θ_o , but is not a necessary condition for the mechanism to work. The mechanism relies instead on the oil sector

using fewer intermediates and importing a smaller share of them, and Table 4 shows that both hold in most of the countries in the sample, many of which do not have a state oil monopoly. What does matter is the share of oil rents that accrues to the government. In a version of the model in which the oil sector is operated by domestic private firms and taxed like the rest of private income, spreads still rise and peak in the same period, but the peak is 0.53 percentage points (against 2.15 in the benchmark).³⁵

4.4 Other drivers

This paper highlights a channel that deteriorates the price schedule of debt following an oil discovery, even before the government increases its borrowing. Subsection 4.3 also shows that the data for several oil-producing economies satisfy a sufficient condition for this deterioration. This Subsection engages with other mechanisms that may drive the response of spreads. Some are implicitly included in the model in a simplified fashion, like the front-loading of future oil income, while others may independently affect the increase in spreads. In either case, it is worth noting that the mechanism presented above depends on parameters that are disciplined directly from the data and, importantly, are not chosen to explain the very response in spreads that this paper studies.

Front-loading future oil revenue.—This is the main driver of the increase in borrowing following discoveries, and is a standard feature of models that are consistent with the permanent-income hypothesis (see for example [Jaimovich and Rebelo \(2008\)](#) and [Jaimovich and Rebelo \(2009\)](#)). Even when borrowing terms deteriorate, an agent (in this case the government) still has incentives to borrow following the news as long as the increase in future income is large enough or as long as the agent is sufficiently impatient. Subsection 4.3 shows that, with identical technologies across sectors, front-loading alone delivers roughly half of the peak increase in spreads.

Overborrowing and political economy frictions.—The model in this paper features a benevolent government, but a notion of “overborrowing” may arise if the government is relatively more impatient than the domestic households due to political economy frictions (see [Hatchondo and](#)

³⁵This version is discussed in 4.5 below, and the difference is fiscal rather than technological. The government collects the income tax rate τ on the oil firm’s profits instead of receiving all of them, so a discovery raises future government revenue by a fraction τ of what it does in the benchmark, and the response of spreads scales roughly with that fraction. Note also that tax revenue of 15.9 percent of GDP is relatively low, and resource-extraction sectors like oil tend to be taxed more heavily than the rest of the economy.

Martinez (2010), Amador (2012), Chatterjee and Eyigungor (2019)). Overborrowing may also arise in an environment with private borrowing and sovereign risk. In such an environment, it is an increase in household debt in response to higher future income that inefficiently increases default risk through a pecuniary externality on sovereign spreads (see Kim and Zhang (2012) and Arce (2024)). The evidence in Section 2, however, points to the government rather than to households. While discoveries raise government consumption, their effect on private consumption is modest. Other political economy frictions may increase the pro-cyclicality of government expenditure following a commodity windfall (see Tornell and Lane (1999)) and raise the volatility of the business cycle (see Aguiar, Amador, and Gopinath (2009)). All these mechanisms would, in turn, increase sovereign spreads if an oil discovery exacerbates them. Exploring them further is beyond the scope of this paper.

(More) complete markets.—One could argue that future oil revenue can be pledged as collateral, allowing the economy to borrow against it without affecting default risk. This, however, would be at odds with the overall low issuance of state-contingent sovereign debt contracts and with the large risk premia carried by the ones that do exist. Roch and Roldán (2023) reconcile this with a theory of state-contingent sovereign debt in an environment in which robust lenders fear model misspecification. In such an environment, bonds indexed to the future revenues from the oil field would also feature high spreads if lenders are uncertain about the stochastic process of future oil revenues. Such uncertainty may arise from the unknown size of the field (Katovich (2024)), from a lack of commitment to oil-production plans (Hamann, Mendez-Vizcaino, Mendoza, and Restrepo-Echavarria (2023)), or from the high volatility of the price of oil.

Dutch disease.—The classic Dutch-disease narrative argues that, upon a commodity windfall, other exporting sectors in the economy contract because of a wealth effect that diverts production factors toward non-traded sectors (see Auty (1993)). This reallocation can be inefficient (hence the “disease”) if the other exporting sectors feature a productivity externality (see Alberola and Benigno (2017)). In such an environment, sovereign spreads would increase following an oil discovery because of the decrease in productivity that would follow the reallocation of production factors toward the oil and non-traded sectors. This would be a complementary mechanism to the one presented in this paper, but it would be one that is almost directly assumed (by assuming the productivity externality) and for which the evidence is not conclusive (Allcott and Keniston

(2018)). [Esquivel \(2025\)](#) studies how the Dutch disease amplifies a pecuniary externality of the sectoral allocation of capital on sovereign risk. In that environment, spreads are inefficiently large because private households fail to internalize that a relatively large non-traded sector increases spreads for the government. In this paper I abstract from capital accumulation for simplicity, but such a mechanism would be complementary to the one presented above.

Investment inflows.—There is also evidence that foreign direct investment (FDI) significantly increases following oil discoveries (see [Toews and Vézina \(2022\)](#) and [Sheng and Zhao \(2024\)](#)). As noted above, the model in this paper abstracts from any form of capital accumulation for simplicity. Having said that, the increase in the fixed factor N_t , which I currently interpret as an increase in the size of the field, could be re-interpreted as an increase in the total production capacity of the field, which combines the natural resource and physical capital. Absent frictions to FDI, such a model would be isomorphic to the benchmark discussed above.³⁶

Official lending.—Unlike the equity inflows just discussed, official lending is a claim on the government. [Horn, Reinhart, and Trebesch \(2021\)](#) document that China has become the largest official creditor in recent years, surpassing multilateral institutions, and oil exporters have outstanding loans from China that range between 2 and 13 percent of their GDP. Some of these loans are related to China’s Belt and Road Initiative, which in the case of the oil industry finances the development of oil fields. Such loans would resemble a version of this model with capital accumulation in the oil sector financed with official debt that may be senior to, and treated more favorably in a restructuring than, the rest of government debt. This configuration does not reverse the mechanism. These loans finance rigs, pipelines and processing plants, not the recurring imported intermediates that Γ measures. Even if they did, they are not cut off by a default in the data. [Arellano and Barreto \(2025\)](#) document that official sovereign debt tends to flow in during default episodes and is more concessional after a default.³⁷ A source of oil-sector external financing that expands during default further lowers Γ . Discoveries in the data, however, do trigger an increase in government consumption and imports, which have to be financed by other type of debt issuance.

³⁶Previous versions of this paper included capital accumulation, both domestic and foreign through FDI. I removed them because the results do not change significantly, so capital adds unnecessary complexity.

³⁷Ecuador defaulted in December 2008 and signed an oil pre-payment facility with PetroChina in July 2009 (Ecuador remained excluded from bond markets until 2014). More broadly, the data in [Horn, Reinhart, and Trebesch \(2021\)](#) show Chinese official lending to Ecuador and Venezuela expanding while both faced exclusion or limited access to bond markets.

Moreover, an increase in official senior debt may further deteriorate the market value of privately-held dollar-denominated bonds, which are the instruments included in the EMBI data. Both the composition of the financing need and the seniority of official debt would be consistent with the evidence from Section 2 and complementary to the main mechanism in the model. Finally, [Horn, Reinhart, and Trebesch \(2021\)](#) also document that some of these official loans are treated as advance payments by China's state-owned oil importers. These instruments are akin to selling future oil rents in exchange for resources in the present, which is one of the cases that I explore in the following subsection.

These mechanisms can be told apart by the margin on which they operate. Most of them raise spreads through borrowing alone, which moves the government along the price schedule. The mechanism in this paper shifts the schedule itself, as do the Dutch-disease externalities and the official lending discussed above. Thus, as long as the sufficient condition $\Gamma < 1$ is satisfied in the data, the mechanism in this paper operates alongside other drivers of the increase in spreads following a discovery.

4.5 Welfare analysis

The model above reconciles the puzzling findings from Section 2 by noting that the higher default risk is driven by how default affects the oil sector differently. From a welfare point of view there is a tension between higher output from a giant oil discovery and higher default risk. Is it worth it to find and exploit giant oil fields?

I compute welfare gains of an oil discovery in terms of consumption equivalent units. Since households value both private consumption (net of labor disutility) and government consumption, I define welfare gains of a discovery in t as the percentage increase χ_t in both, such that households are indifferent between not discovering a giant field and discovering it. Since preferences are homothetic and $\sigma = \sigma^G$, we can write welfare gains in terms of the state and value functions

$$\chi(b, z) = 100 * \left[\left(\frac{V(b, z, 0)}{V(b, z, -1)} \right)^{\frac{1}{1-\sigma}} - 1 \right],$$

where $n = -1$ indicates that the economy has not discovered an oil field and $n = 0$ indicates news of an oil discovery.

Table 5: Average welfare gains of giant oil discoveries

	benchmark	oil firm operated by domestic private sector	sell oil rents from giant field to foreigners
	(1)	(2)	(3)
welfare gains	7.6	2.4	6.2

Welfare gains are computed in consumption equivalent units. The numbers are the averages of 1,000 discoveries randomly drawn from the ergodic distribution.

Table 5 presents the average welfare gains of oil discoveries in three versions of the model. Column (1) corresponds to the benchmark and shows that there are sizable welfare gains from oil discoveries, despite the increase in spreads. The gains of 7.6 percent, however, contrast with the average 20 percent increase in government consumption shown in Figure 6. The difference is explained by the higher default risk and by a composition effect because the government cannot increase private consumption—while the compensating variation does by construction. Column (2) corresponds to a version of the model in which the oil firm is operated by the domestic private sector and taxed in the same way as the rest of private income. In this case, the budget constraint of the government in good standing is

$$g_t + \gamma b_t \leq \tau [w_t L_t + \pi_{ft} + \pi_{dt} + \pi_{ot}] + q_t i_{b,t}$$

and in default it is

$$g_t \leq \tau [w_t L_t + \pi_{ft} + \pi_{dt} + \pi_{ot}].$$

Interestingly, welfare gains of oil discoveries are smaller in this case. This is because the government has access to only 15.9 percent of the windfall, so the absolute quantity of resources that can be front-loaded is smaller. Two simplifying assumptions lie behind this: the tax rate is fixed, and the government does not have access to the aggregate resource constraint via lump-sum transfers. If the government could tax oil profits at a separate rate, or make lump-sum transfers, then privatization would not preclude the government from capturing the entirety of the windfall, as in the benchmark.³⁸ These results would of course be different if the government were assumed not to

³⁸There is also a composition effect. Utility is separable, $\sigma = \sigma_g$, and $\psi_g = 1$. These assumptions, along with the relatively small τ , make g the scarcer good with a higher marginal utility. The entire windfall accrues to g in the benchmark, while most of it accrues to c with a domestic private oil firm. Setting ψ_g so that the government is indifferent between the two goods would eliminate this effect. This symmetry was assumed for simplicity, and properly pinning down those two parameters and the separability of utility is well beyond the scope of this paper.

be benevolent. The government could be more impatient than the households or have a relatively higher weight on government consumption $\psi_g > 1$. In either case, privatizing the oil sector would, by assumption, give more agency to the households by removing tools from a non-benevolent government, unambiguously increasing welfare regardless of the details of the oil sector. While these considerations are interesting, they are beyond the scope of this paper.

Finally, Column (3) presents the welfare gains from selling oil rents from the giant oil discovery to foreigners. Define total oil rents as $R_{ot} = \alpha_N y_{ot}$, the rents that corresponds to the giant oil field are

$$R_{ot}^* = \alpha_N \frac{y_{ot}}{N_H} [N_H - N_L],$$

if foreigners discount the future at a rate equal to the risk-free rate r^* then we can approximate the present value of these rents as

$$v = \left(\frac{1}{1+r} \right)^{\text{Twait}} \frac{1+r^*}{r^* + \pi_{\text{ex}}} R_{oss}^*$$

where R_{oss}^* assumes $z_t = 1$ during the life of the giant field for simplicity. The welfare gains in this case are smaller than in the benchmark. Selling the rents changes the budget constraint of the government in two ways: it receives v in the period of the discovery, and it gives up the flows R_{ot}^* in every period in which the giant field is active, whether it repays or defaults. Following [Aguilar, Amador, and Fourakis \(2020\)](#), I decompose the gains from selling by turning these two changes on one at a time. Let ζ denote the gains from selling the rents, in consumption equivalent units and conditional on a discovery, so that

$$1 + \zeta = \left[\frac{V^S(b, z, 0)}{V^K(b, z, 0)} \right]^{\frac{1}{1-\sigma}} = \underbrace{\left[\frac{V^R}{V^K} \right]^{\frac{1}{1-\sigma}}}_{1+\zeta_R} \times \underbrace{\left[\frac{V^F}{V^R} \right]^{\frac{1}{1-\sigma}}}_{1+\zeta_D} \times \underbrace{\left[\frac{V^S}{V^F} \right]^{\frac{1}{1-\sigma}}}_{1+\zeta_W}, \quad (36)$$

where all four value functions are evaluated at the same state in the period of the discovery. V^K is the value function of the benchmark economy. V^R is the value function of an economy in which the government gives up the flows R_{ot}^* in the periods in which it repays and receives nothing in exchange. V^F is the value function of an economy in which it gives up those flows in every period, including the periods it spends in default (the giant field is still discovered and produces, only the

giant field's rents accrue abroad). V^S is the value function of the economy in Column (3), which sells the discovered field for v in the discovery year.

The government gives up a stream of rents that it values at 8.0 percent of consumption— $\zeta_R = -2.7$ in the periods in which it repays and $\zeta_D = -5.4$ in the periods it spends in default—in exchange for a payment that it values at $\zeta_W = 7.2$ percent.³⁹ Selling therefore lowers welfare, by $\zeta = -1.3$ percent, because the government values the rents it gives up by more than the risk-free price that foreign buyers pay for them. The government's relative impatience would make the sale attractive on its own, but by selling the field the government foregoes additional income during default, when the marginal utility of consumption is the largest. This is why the loss in default is twice as large as the one in repayment, despite the improvement in borrowing terms that a worse default value would imply (keeping these rents in default makes defaulting more attractive, which raises spreads). Welfare falls in this second step because the loss of the additional resources during default with a large field is not compensated by the gain of lower spreads in good times. Oil rents are thus worth disproportionately more to the government in default, which is the same property that makes spreads increase after a discovery; here it appears as what makes selling those rents expensive, rather than as what makes selling them attractive.⁴⁰

Finally, I evaluate how much of the welfare gains of a discovery is accounted for by the fact that oil is relatively insulated from default. The answer is little. In the two economies of Subsection 4.3 that reduce the sectoral differences—the economy with $\theta_o = \theta_f$ and the one with identical technologies—welfare gains of a discovery are 7.65 and 6.88 percent of consumption, against 7.62 percent in the benchmark. The same sectoral differences remove the shift in the price schedule entirely, so the composition of the default penalty matters a great deal for the terms at which the government borrows and comparatively little for the value of the discovery itself.

³⁹All three terms are averages over the same 1,000 discoveries drawn from the ergodic distribution used in Table 5.

⁴⁰This is the channel through which the accumulation of reserves raises sovereign risk in Hamann, Mendez-Vizcaino, Mendoza, and Restrepo-Echavarria (2023). In this decomposition, ζ_D measures what the giant field's rents are worth to the government in autarky, and it is twice ζ_R . In my model, that gap follows from the differences in technology and financing needs across sectors rather than from a choice over extraction.

5 Conclusion

In this paper, I documented a puzzling response of sovereign spreads to giant oil field discoveries: spreads increase significantly even though oil discoveries are news of higher future income. Spreads are countercyclical in the data and sovereign default theory has built on the assumption that the real costs of default are increasing in income to explain this regularity. To reconcile these findings with existing theory I developed a model in which the oil sector is endogenously less affected by a sovereign default than the rest of the economy. Critically, it is the oil sector's relatively lower dependence on imported intermediates and on working capital that drives this lower exposure. I discipline these differences using Mexico's Input-Output Matrix and financial statements from the largest oil firm in the country. The model successfully generates the joint responses to giant oil discoveries without sacrificing its ability to match business cycle regularities from the data. Moreover, I show that this mechanism is summarized by a sufficient statistic that compares how exposed the two sectors are to the cost of default and that can be computed from an input-output table alone. I compute it for a cross-section of oil-producing economies and find that the condition under which a discovery worsens borrowing terms is satisfied for most of them, so the mechanism does not rest on the details of the Mexican calibration.

Despite the increase in default risk that they generate, there are significant welfare gains from oil discoveries. Interestingly, these gains are smaller if the domestic private sector benefits directly from oil rents and remain large if the government sells the additional oil rents to foreigners. This is because selling the rents reduces default incentives for future governments, who would otherwise take advantage of higher oil rents and lower costs of defaulting to increase borrowing and default risk. Studying how fiscal rules may be adapted to tame these incentives during oil windfalls is an exciting avenue for future research.

References

- Aguiar, Mark and Manuel Amador. 2014. “Sovereign Debt.” In *Handbook of International Economics*, vol. 4. 647–687. 29
- Aguiar, Mark, Manuel Amador, and Stelios Fourakis. 2020. “On the Welfare Losses from External Sovereign Borrowing.” *IMF Economic Review* 68 (1):163–194. 43
- Aguiar, Mark, Manuel Amador, and Gita Gopinath. 2009. “Investment Cycles and Sovereign Debt Overhang.” *Review of Economic Studies* 76 (1):1–31. 39
- Aguiar, Mark, Satyajit Chatterjee, Harold L. Cole, and Zachary Stangebye. 2016. “Quantitative Models of Sovereign Debt Crises.” In *Handbook of Macroeconomics*. 24
- Aguiar, Mark and Gita Gopinath. 2006. “Defaultable Debt, Interest Rates and the Current Account.” *Journal of International Economics* 69 (1):64–83. 3
- . 2007. “Emerging Market Business Cycles: The Cycle Is the Trend.” *Journal of Political Economy* 115 (1):71–102. 24
- Alberola, Enrique and Gianluca Benigno. 2017. “Revisiting the commodity curse: A financial perspective.” *Journal of International Economics* 108:87–106. 39
- Allcott, Hunt and Daniel Keniston. 2018. “Dutch Disease or Agglomeration? The Local Economic Effects of Natural Resource Booms in Modern America.” *The Review of Economic Studies* 85 (2):695–731. 39
- Almeida, Victor, Carlos Esquivel, Timothy J. Kehoe, and Juan Pablo Nicolini. 2024. “Default and Interest Rate Shocks: Renegotiation Matters.” Working Paper 806, Federal Reserve Bank of Minneapolis. 19
- Amador, Manuel. 2012. “Sovereign Debt and The Tragedy of the Commons.” Working Paper. 39
- Arce, Fernando. 2024. “Private Overborrowing under Sovereign Risk.” *Journal of Political Economy Macroeconomics* 2 (4):553–601. 39

- Arellano, Cristina. 2008. “Default Risk and Income Fluctuations in Emerging Economies.” *American Economic Review* 98 (3):690–712. 1, 3, 10, 11, 12, 22, 28, 29, 62
- Arellano, Cristina, Yan Bai, and Luigi Bocola. 2017. “Sovereign Default Risk and Firm Heterogeneity.” Working Paper 23314, National Bureau of Economic Research. 2
- Arellano, Cristina and Leonardo Barreto. 2025. “Official Sovereign Debt.” Staff Report 678, Federal Reserve Bank of Minneapolis. 40
- Arezki, Rabah, Valerie A. Ramey, and Liugang Sheng. 2017. “News Shocks in Open Economies: Evidence From Giant Oil Discoveries.” *Quarterly Journal of Economics* :103–155. 1, 5, 6, 7, 8, 9, 10, 52, 56, 57, 61
- Arteta, Carlos and Galina Hale. 2008. “Sovereign debt crises and credit to the private sector.” *Journal of International Economics* 74 (1):53–69. 16
- Asonuma, Tamon, Dirk Niepelt, and Romain Rancière. 2023. “Sovereign bond prices, haircuts and maturity.” *Journal of International Economics* 140:103689. 25
- Auty, Richard. 1993. *Sustaining Development in Mineral Economies*. London: Routledge. 39
- Bajzik, Josef, Tomas Havranek, Zuzana Irsova, and Jiri Schwarz. 2020. “Estimating the Armington elasticity: The importance of study design and publication bias.” *Journal of International Economics* 127:103383. 15, 63
- Beaudry, Paul and Franck Portier. 2014. “News-Driven Business Cycles: Insights and Callenges.” *Journal of Economic Literature* 52 (4):993–1074. 5
- Benjamin, David and Mark L.J. Wright. 2009. “Recovery Before Redemption: A Theory of Delays in Sovereign Debt Renegotiations.” Working paper, Available at SSRN. 19
- Bianchi, Javier, Juan Carlos Hatchondo, and Leonardo Martinez. 2018. “International Reserves and Rollover Risk.” *American Economic Review* 108 (9):2629–2670. 24
- Blonigen, Bruce A. and Wesley W. Wilson. 1999. “Explaining Armington: What Determines Substitutability between Home and Foreign Goods?” *Canadian Journal of Economics* 32 (1):1–21. 63

- Bocola, Luigi. 2016. “The Pass-Through of Sovereign Risk.” *Journal of Political Economy* 124 (4):879–926. 2, 17
- Boehm, Christoph E., Andrei A. Levchenko, and Nitya Pandalai-Nayar. 2023. “The Long and Short (Run) of Trade Elasticities.” *American Economic Review* 113 (4):861–905. 63
- Bornstein, Gideon, Per Krusell, and Sergio Rebelo. 2022. “A World Equilibrium Model of the Oil Market.” *The Review of Economic Studies* 90 (1):132–164. 18
- Broda, Christian and David E. Weinstein. 2006. “Globalization and the Gains from Variety.” *The Quarterly Journal of Economics* 121 (2):541–585. 64
- Chatterjee, Satyajit and Burcu Eyigungor. 2012. “Maturity, Indebtedness, and Default Risk.” *American Economic Review* 102 (6):2674–2699. 19
- . 2019. “Endogenous political turnover and fluctuations in sovereign default risk.” *Journal of International Economics* 117:37–50. 39
- Dickey, David A. and Wayne A. Fuller. 1979. “Distribution of the Estimators for Autoregressive Time Series With a Unit Root.” *Journal of the American Statistical Association* 74 (366):427–431. 8
- Drechsel, Thomas and Silvana Tenreyro. 2018. “Commodity booms and busts in emerging economies.” *Journal of International Economics* 112:200–218. 4, 18
- Driscoll, John C. and Aart C. Kraay. 1998. “Consistent Covariance Matrix Estimation with Spatially Dependent Panel Data.” *Review of Economics and Statistics* 80:549–560. 9, 10, 53, 55, 56, 57, 58, 59, 60, 61
- Dvorkin, Maximiliano, Juan M. Sanchez, Horacio Sapriza, and Emircan Yurdagul. 2020. “News, sovereign debt maturity, and default risk.” *Journal of International Economics* 126:103352. 5
- . 2021. “Sovereign Debt Restructurings.” *American Economic Journal: Macroeconomics* 13 (2):26–77. 19
- Eaton, Jonathan and Mark Gersovitz. 1981. “Debt with Potential Repudiation: Theoretical and Empirical Analysis.” *The Review of Economic Studies* 48 (2):289–309. 3

- Esquivel, Carlos. 2025. “Sovereign risk and Dutch disease.” *Journal of Monetary Economics* 149:103663. 40
- Gelos, R. Gaston, Ratna Sahay, and Guido Sandleirs. 2011. “Sovereign borrowing by developing economies: What determines market access?” *Journal of International Economics* 83:243–254. 25
- Greenwood, Jeremy, Zvi Hercowitz, and Gregory W. Huffman. 1988. “Investment, Capacity Utilization, and the Real Business Cycle.” *American Economic Review* 78 (3):402–417. 18
- Hamann, Franz, Juan Camilo Mendez-Vizcaino, Enrique G Mendoza, and Paulina Restrepo-Echavarria. 2023. “Natural Resources and Sovereign Risk in Emerging Economies: A Curse and a Blessing.” Working Paper 31058, National Bureau of Economic Research. 3, 9, 39, 44, 52, 53
- Hatchondo, Juan Carlos and Leonardo Martinez. 2009. “Long-Duration Bonds and Sovereign Defaults.” *Journal of International Economics* 79 (1):117–125. 19
- . 2010. “The Politics of Sovereign Defaults.” *Federal Reserve Bank of Richmond Economic Quarterly* 96 (3):291–317. 38
- Hatchondo, Juan Carlos, Leonardo Martinez, and Horacio Sapriza. 2010. “Quantitative properties of sovereign default models: Solution methods matter.” *Review of Economic Dynamics* 13 (4):919–933. 24
- Horn, Myron K. 2014. “Giant Oil and Gas Fields of the World.” <https://edx.netl.doe.gov/dataset/aapg-datapages-giant-oil-and-gas-fields-of-the-world>. 5, 6
- Horn, Sebastian, Carmen M. Reinhart, and Christoph Trebesch. 2021. “China’s overseas lending.” *Journal of International Economics* 133:103539. 9, 40, 41
- Im, Kyung So, M.Hashem Pesaran, and Yongcheol Shin. 2003. “Testing for unit roots in heterogeneous panels.” *Journal of Econometrics* 115 (1):53–74. 8

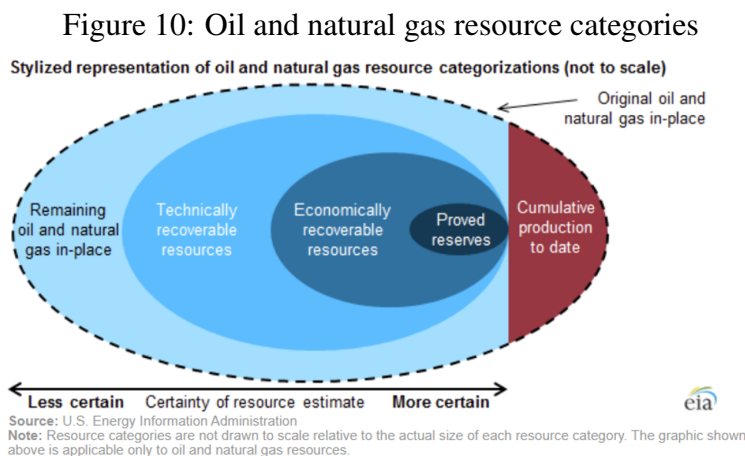
- Jaimovich, Nir and Sergio Rebelo. 2008. “News and Business Cycles in Open Economies.” *Journal of Money, Credit and Banking* 40 (8):1699–1711. 5, 38
- . 2009. “Can News about the Future Drive the Business Cycle?” *American Economic Review* 99 (4):1097–1118. 38
- Karabarbounis, Loukas and Brent Neiman. 2013. “The Global Decline of the Labor Share.” *The Quarterly Journal of Economics* 129 (1):61–103. 26
- Katovich, Erik S. 2024. “Winning and losing the resource lottery: Governance after uncertain oil discoveries.” *Journal of Development Economics* 166:103204. 39
- Kim, Yun Jung and Jing Zhang. 2012. “Decentralized borrowing and centralized default.” *Journal of International Economics* 88 (1):121–133. 39
- Kohlscheen, Emanuel and Stephen A. O’Connell. 2007. “Trade Credit, International Reserves and Sovereign Debt.” Warwick Economic Research Papers 833, University of Warwick, Department of Economics. 17
- Laeven, Luc and Fabian Valencia. 2018. “Systemic Banking Crises Revisited.” Working Paper WP/18/206, International Monetary Fund. 2, 17
- Mendoza, Enrique. 1991. “Real Buisness Cycles in a Small Open Economy.” *American Economic Review* 81 (4):797–818. 24
- Mendoza, Enrique G. and Vivian Z. Yue. 2012. “A General Equilibrium Model of Sovereign Default and Buisness Cycles.” *Quarterly Journal of Economics* 127:889–946. 2, 3, 14, 16, 17, 22, 24, 25, 26, 27, 28, 29, 31, 33, 63, 64
- Neumeyer, Pablo A. and Fabrizio Perri. 2005. “Business cycles in emerging economies: the role of interest rates.” *Journal of Monetary Economics* 52:345–380. 24
- Reinhart, Carmen M. 2010. “This Time is Different Chartbook: Country Histories on Debt, Default, and Financial Crises.” Working Paper 15815, National Bureau of Economic Research. 17

- Reinhart, Carmen M. and Kenneth S. Rogoff. 2011. “From Financial Crash to Debt Crisis.” *American Economic Review* 101 (5):1676–1706. 2, 17
- Roch, Francisco and Francisco Roldán. 2023. “Uncertainty Premia, Sovereign Default Risk, and State-Contingent Debt.” *Journal of Political Economy Macroeconomics* 1 (2):334–370. 39
- Schmitt-Grohe, Stephanie and Martín Uribe. 2007. “Optimal simple and implementable monetary and fiscal rules.” *Journal of Monetary Economics* 54 (6):1702–1725. 28
- Sheng, Liugang and Hongyan Zhao. 2024. “Oil Shocks, External Adjustment, and Country Portfolio.” *Journal of Money, Credit and Banking* 56 (7):1705–1736. 10, 40
- Toews, Gerhard and Pierre-Louis Vézina. 2022. “Resource Discoveries, FDI Bonanzas, and Local Multipliers: Evidence from Mozambique.” *The Review of Economics and Statistics* 104 (5):1046–1058. 10, 40
- Tornell, Aaron and Philip R. Lane. 1999. “The Voracity Effect.” *American Economic Review* 89 (1):22–46. 39
- Wright, Mark L. J. 2006. “Private capital flows, capital controls, and default risk.” *Journal of International Economics* 69 (1):120–149. 17

A Additional empirical results

A.1 Controlling for oil reserves

As documented by [Hamann, Mendez-Vizcaino, Mendoza, and Restrepo-Echavarria \(2023\)](#), the dynamics of proved oil reserves have a significant impact on the evolution of credit worthiness of emerging economies who are oil exporters. In order to understand my findings in light of their results it is important to note a conceptual distinction between proved oil reserves and URR. There is a range of categories to measure oil reserves. Figure 10 shows a conceptual diagram from the U.S. Energy Information Administration that illustrates the differences between these categories.

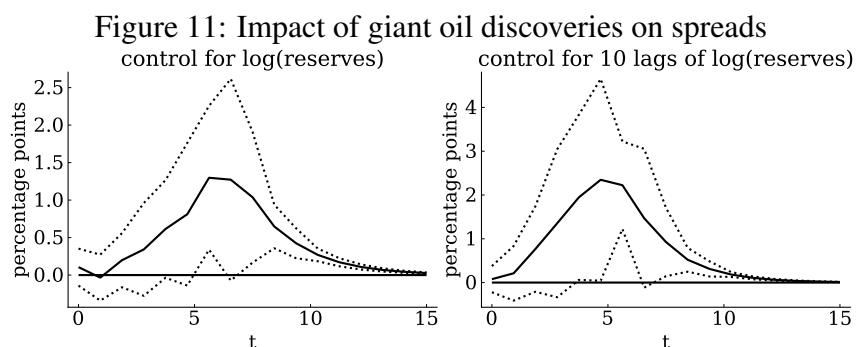


Each category implies a different level of uncertainty, where the most certain measure is proved reserves and the most uncertain is remaining oil and natural gas in-place. Oil and gas in-place refers to the total amount of resources within a geological formation. Technically recoverable resources includes oil and gas that can be produced based on current technology.⁴¹ This is the estimate of URR that [Arezki, Ramey, and Sheng \(2017\)](#) use to construct the NPV of oil fields, which can be interpreted as the amount of oil in a field that is physically feasible to extract. Economically recoverable resources (ERR) are all URR that can be profitably produced given economic conditions (like the price of oil and variable costs of production) at the time of measurement. Finally, proved oil reserves require a higher standard of certainty to be considered profitably and physically recoverable. As ERR, proved reserves shrink and grow as the prices of oil and extraction inputs vary. URR do not.

⁴¹Geophysical characteristics of rocks, as well as physical properties of hydrocarbons (such as viscosity) prevent technology from producing the entirety of the ultimately recoverable reserves.

It is crucial to note that, by definition, the resources contained in giant oil field discoveries are not included in the measure of proved oil reserves at the time of the discovery. Instead, the oil in a field is gradually added to proved reserves once drilling starts and new information is collected about its feasibility and profitability.

[Hamann, Mendez-Vizcaino, Mendoza, and Restrepo-Echavarria \(2023\)](#) document how marginal changes in proved oil reserves impact the credit worthiness of oil exporting countries, identifying both long and short-run effects. The shocks these authors identify are driven by international economic conditions (like oil prices) and by endogenous extraction decisions, both of which are the main source of variation in proved oil reserves. There are three important differences between [Hamann, Mendez-Vizcaino, Mendoza, and Restrepo-Echavarria \(2023\)](#) and the work presented in this paper. The first has to do with the magnitude of the shocks at hand. By definition, the size of year-to-year changes in proved reserves is dwarfed by the size of giant oil discoveries. The second has to do with the fact that newly discovered giant oil fields cannot be immediately exploited; instead, they require a substantial amount of investment through several years in order to become productive. In contrast, proven reserves can be more easily exploited within shorter periods of time. The third has to do with the source of the variation. Changes in proved reserves respond to oil prices and to extraction decisions, so they measure how sovereign risk moves with a stock that the government manages. The timing of a giant discovery is not chosen, which allows the estimates in Section 2 to be read as the response of spreads to news about future output alone. These differences imply that discoveries may affect sovereign interest rate spreads in a way that marginal changes in proved reserves do not.



Impulse response to an oil discovery with net present value equal to 4.5 percent of GDP, which is the median size of discoveries in the sample. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

Figure 11 shows the dynamic response of the spreads following a discovery of median size. The left panel controls for the natural logarithm of contemporaneous proved reserves and the right panel controls for this and ten lags. The results are very similar to the benchmark results that do not control for reserves.

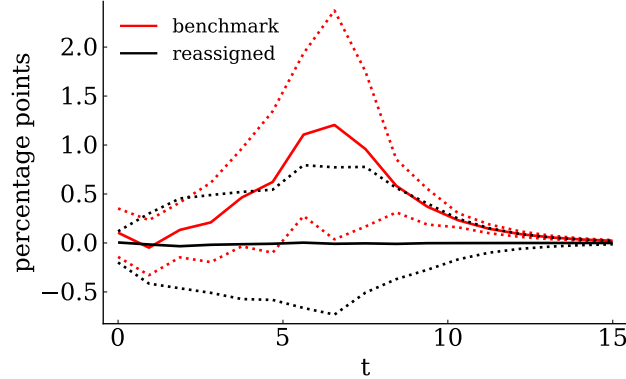
A.2 Placebo discoveries

A potential identification concern is that spreads in the countries where discoveries happen may be correlated to unobserved policy changes or global oil-market dynamics. To show that the estimated effect is directly tied to discovering a giant oil field, I now perform placebo regressions in which I randomly treat countries in the sample with oil discoveries. Specifically, for each placebo regression I reassign the fifteen discovery histories observed in the data to fifteen countries selected at random.⁴² The sizes of the discoveries and the years in which they occur are the actual ones, and only the country that received them changes.⁴³ For each random reassignment, I estimate equation (2) for spreads and compute the dynamic response as in Figure 2. I repeat this exercise 1,000 times. Figure 12 presents the median (solid black line) and the fifth and ninety-fifth percentiles (dashed black lines) of the response across placebo regressions. The estimated response from the actual data and its 90-percent confidence intervals from Figure 2 are in red.

⁴²Out of the 37 countries with EMBI spreads data, 15 had at least one oil discovery.

⁴³With this formulation I am considering having oil discoveries as the treatment, and the quantity and sizes of discoveries as the intensity of treatment. This keeps the distribution of sizes of discoveries the same across regressions, which makes it easier to compare impulse-response functions.

Figure 12: Response of spreads with discovery histories reassigned



Impulse response to an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The benchmark response of Figure 2 is in red, with dotted lines indicating 90 percent confidence intervals based on a Driscoll and Kraay (1998) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering. The black lines are the median and the fifth and ninety-fifth percentiles of the response across 1,000 replications in which the discovery histories are reassigned at random among the 37 EMBI countries.

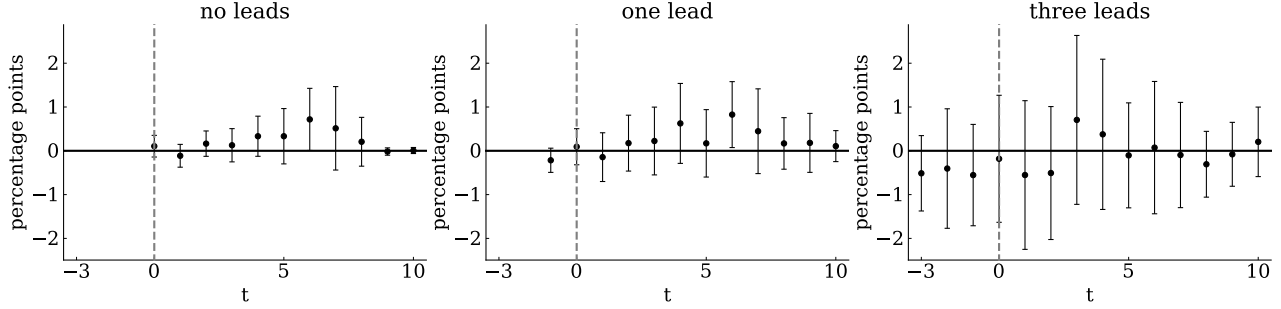
Figure 12 suggests that the response of spreads is specific to the countries that actually had oil discoveries. Randomly reassigned responses are centered around zero at every horizon. The median never reaches 0.04 percentage points in absolute value, which is why the solid black line appears to be a horizontal line at zero. The ninety-fifth percentile peaks at 0.8, while the benchmark estimation peaks at 1.2 in the sixth year. Fewer than 7 percent of random reassignments produce a response as large as the benchmark, whether this magnitude is measured by the peak of the response, by the response in year 6 (when the benchmark peaks), or cumulatively over the 15-year horizon.

A.3 Pre-trends

I now estimate equation (2) adding leads of the NPV of a discovery to test for pre-trends in the spreads that may not be captured by the country-specific quadratic trends. The sum over s in equation (2) now starts at $s = -1$ and at $s = -3$ rather than at $s = 0$. Figure 13 presents the estimated coefficients ψ_s from the benchmark specification for spreads with no leads (left panel) and from these two specifications (middle and right panels), scaled by the median size of discoveries. The figure presents the coefficients rather than the impulse-response functions because what is at issue in this case is whether the coefficients on the leads are zero, while the impulse response cumulates

each of them through the estimated persistence of spreads ρ .

Figure 13: Estimated coefficients with leads of discovery size

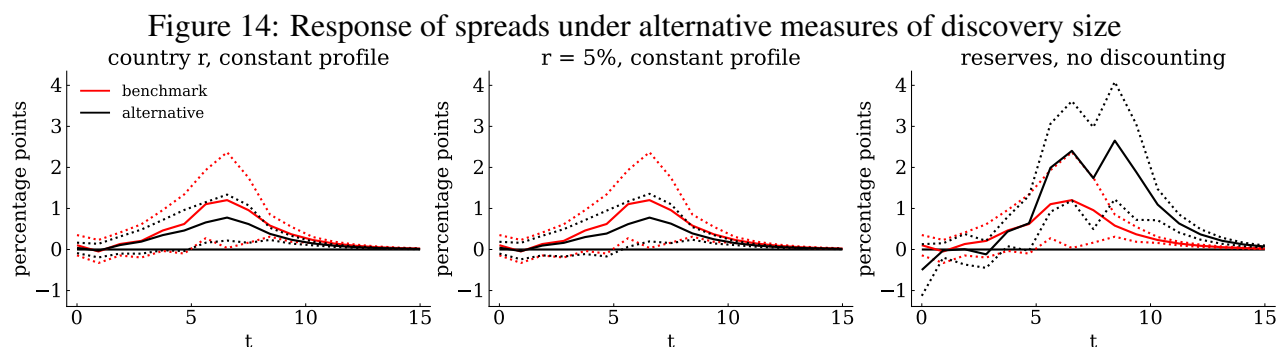


Estimated coefficients ψ_s for an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The bars indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

There is no evidence of a pre-trend in spreads. With one lead ψ_{-1} is -0.216 with a 90 percent confidence interval of $[-0.49, 0.06]$, the tightest interval in that estimation. Moreover, the lags do not change much relative to the benchmark and the response still peaks in the sixth year after the discovery with a similar shape as the impulse-response function in [Figure 2](#). With three leads none of the lead coefficients is distinguishable from zero. All confidence intervals widen because each discovery has to identify fourteen coefficients rather than eleven, but the point estimates give no indication that spreads were rising before a discovery was announced.

A.4 Alternative measures of discovery size

I now estimate the response of spreads to discoveries using alternative measures of size that do not rely on country-specific discount rates. [Figure 14](#) presents impulse-response functions from the estimates of [equation \(2\)](#) using three alternative measures for the size of a discovery. The first uses the same risk-adjusted rates r_i but replaces the estimated production profile in [Arezki, Ramey, and Sheng \(2017\)](#), which features higher production in early years, with a constant one in which production is uniform between the sixth and the twenty-fifth year after the discovery. The second applies a common discount rate of $r = 0.05$ to that same constant profile so that the comparison between the first two isolates the role of the discount rate. The third uses the URR of the field in millions of barrels, which involves no discounting at all. Each panel presents the response under the alternative measure in black and the benchmark response of [Figure 2](#) in red.

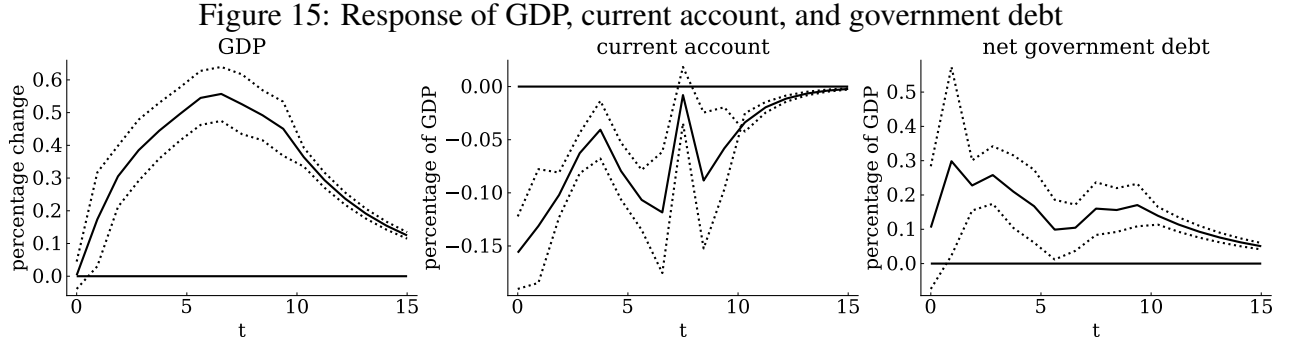


Impulse response to an oil discovery of median size under each measure: 4.5 percent of GDP for the benchmark (red), 3.1 percent for the case with risk-adjusted rates and a constant production profile (left), 4.7 percent for a common discount rate of 5 percent (middle), and 1,000 million barrels of URR for the size in volume of oil (right). The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

The results in Subsection 2.3 are robust to these alternative measures. Spreads increase after a discovery in all three cases, with a hump-shaped response that peaks around the time when production begins and is significantly different from zero at its peak. Holding the production profile fixed, replacing the risk-adjusted rates with a common discount rate of 5 percent moves the peak response from 0.778 to 0.777 percentage points, so the differentiated discounting of future oil revenue plays virtually no role in the estimated response. Measuring the size of a discovery in volume of oil yields a peak response of 2.7 percentage points, more than twice as large as the benchmark, which suggests that using the risk-adjusted measure of [Arezki, Ramey, and Sheng \(2017\)](#) is conservative.

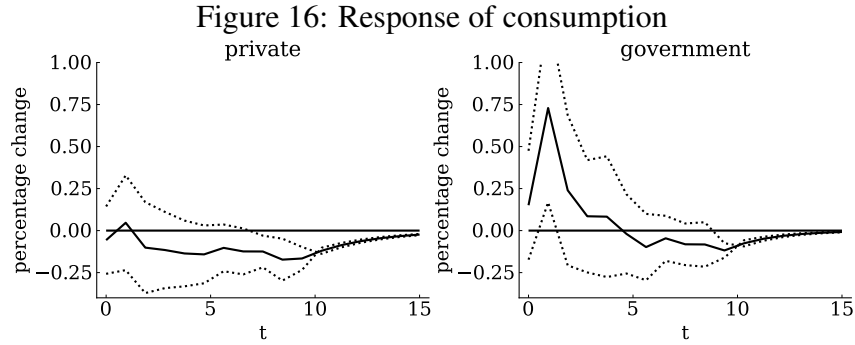
A.5 Estimates with restricted sample

Figure 15 presents the responses of GDP, the current account, and government debt by restricting the sample to the 37 emerging economies in the EMBI.



Impulse response to an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

As discussed in the main text, GDP increases and peaks at around the time when the oil field becomes productive, the economies run a current account deficit and government debt increases. Interestingly, GDP starts increasing right after news arrives and the current account reversal is less pronounced. These interactions with production in the rest of the economy are interesting on their own and beyond the scope of this paper.



Impulse response to an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors, which yields standard error estimates that are robust to general forms of spatial and temporal clustering.

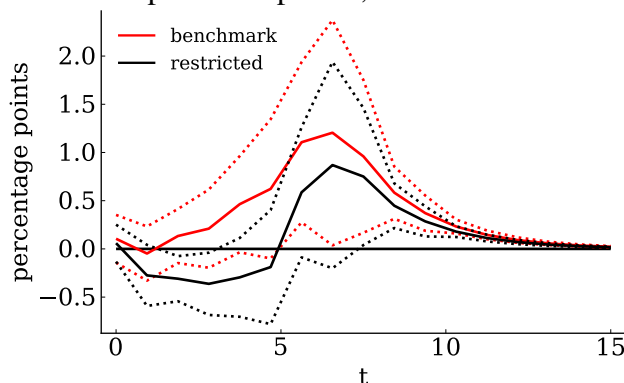
Figure 16 presents the responses of private and public consumption for the restricted sample. As is the case with the results presented in the main text, the response of public consumption is larger than that of private consumption.

A.6 Discoveries before the increase in Chinese official lending

Official lending from China to oil producers expanded sharply after 2008 and part of it financed oil infrastructure, as discussed in Subsection 4.4. If that lending, rather than the mechanism in this paper, were behind the response of spreads to discoveries, then the response should substantially

weaken once the discoveries that could have been financed this way are removed. Figure 17 re-estimates equation (2) setting the size of the nine discoveries that occur in 2009 or later to zero. The 52 remaining discoveries carry 88.5 percent of the total net present value of discoveries in the sample.

Figure 17: Response of spreads, discoveries before 2009



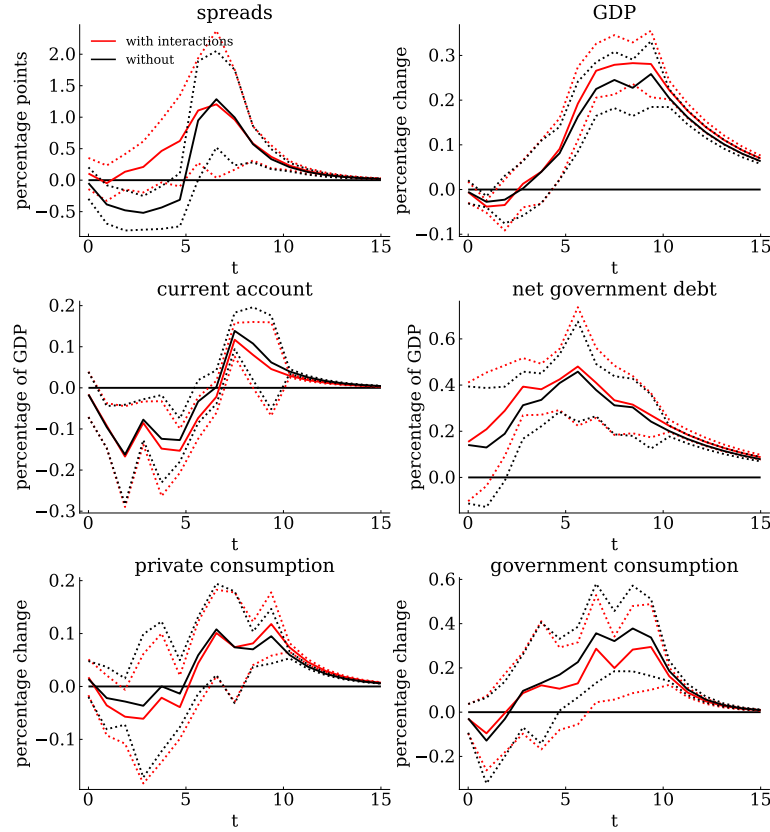
Impulse response to an oil discovery with net present value equal to the median size of 4.5 percent of GDP. The benchmark line reproduces Figure 2. The restricted line sets the size of the nine discoveries that occur in 2009 or later to zero. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors.

The response keeps its shape and its timing, and it peaks at the same horizon as the benchmark. The estimated peak is slightly lower than in the benchmark, but each estimate lies well within the other's confidence interval.

A.7 Estimates without the oil-price interactions

Equation (2) includes interactions of the discovery indicator with the price of oil, which allow the response to the price of oil to differ for countries that recently made a discovery. This appendix re-estimates every outcome without them. Figure 18 presents the two sets of estimates side by side.

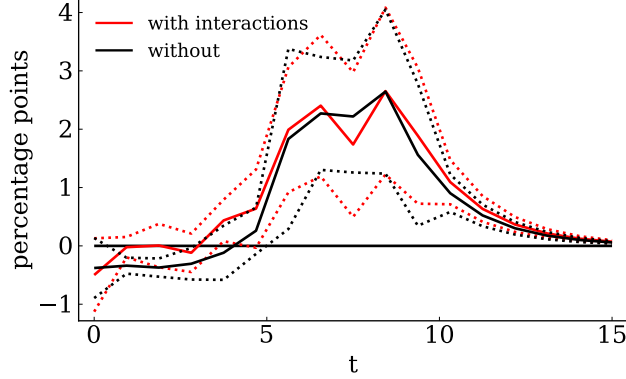
Figure 18: Responses with and without the oil-price interactions



Each panel is the regression behind Figures 2, 3 and 4, estimated with the interactions of the discovery indicator with the log price of oil in red and without them in black. All responses are scaled by a discovery of the median size of 4.5 percent of GDP. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors.

Spreads are the only outcome for which the specification matters. The increase is present in both specifications and peaks in the same year. The two differ in the years right after the discovery, where the response estimated without the interactions is negative. This estimated decrease is likely to be biased downward, because the net present value of a discovery is positively correlated with the price of oil by construction, and high oil prices are in turn associated with low spreads in oil-producing economies. Figure 19 illustrates this by making the same comparison using ultimately recoverable reserves, a measure of the physical size of the field that does not embed the price of oil, instead of the net present value.

Figure 19: Response of spreads using reserves with and without the oil-price interactions



The responses correspond to the regression behind Figure 2 estimated with ultimately recoverable reserves (URR) as the measure of size. The red line includes interactions of the discovery indicator with the log price of oil, and the black excludes them. Both responses are scaled by a discovery of the median size of 1,000 million barrels of URR. The dotted lines indicate 90 percent confidence intervals based on a [Driscoll and Kraay \(1998\)](#) estimation of standard errors.

The interaction controls barely matter in this case, as they do for the other five outcomes in Figure 18. I keep the specification with the interactions and the NPV measure in the main text because the peak increase in spreads is the smallest of the three, 1.21 percentage points against 1.29 without the interactions and 2.7 under reserves, which is more conservative, and to be consistent with [Arezki, Ramey, and Sheng \(2017\)](#).

B Canonical model with oil discoveries

Recursive formulation.—The state of the economy is (b, z, n) . The value of the government at the beginning of a period in good financial standing is

$$V(b, z, n) = \max_{d \in \{0, 1\}} \{dV^D(z, n) + (1 - d)V^P(b, z, n)\}, \quad (37)$$

where the value of repayment is

$$\begin{aligned} V^P(b, z, n) &= \max_{c, b'} \{u(c) + \beta \mathbb{E}[V(b', z', n')]\} \\ \text{s.t.} \quad &c + b \leq z + R^{\text{oil}}(n) + q(b', z, n)b', \end{aligned} \quad (38)$$

where q is the price schedule of debt, and the value of default is

$$\begin{aligned} V^D(z, n) &= u(y^{def}(z, n)) + \beta \phi \mathbb{E}[V(0, z', n')] + \beta(1 - \phi) \mathbb{E}[V^D(z', n')] \\ y^{def}(z, n) &= h_z(z) + h_{oil}(R^{oil}(n)), \end{aligned} \quad (39)$$

where ϕ is the probability of reentry to financial markets. This illustrative model abstracts from recovery, so a default is resolved with reentry at zero debt, which corresponds to $\kappa = 0$ in the notation of Section 3.

Equilibrium.—An equilibrium is value and policy functions and a price schedule q such that: (i) given q , the value and policy functions satisfy equations (37), (38), and (39); and (ii) the price schedule satisfy the lender's no-arbitrage condition:

$$q(b', z, n) = \frac{\mathbb{E}[(1 - d(b', z', n'))]}{1 + r^*},$$

where r^* is the international risk-free rate.

Parameterization and calibration.—The income shock z_t follows an AR(1) process

$$\log z_t = \rho \log z_{t-1} + \sigma_z^2 \varepsilon_t,$$

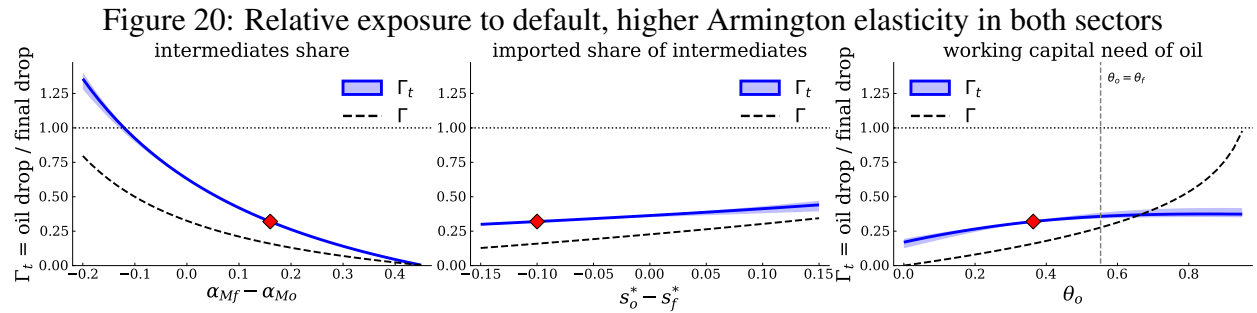
where ε_t follow a standard normal distribution and are iid over time. The utility function is CRRA $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$. Table 6 summarizes the calibration.

Table 6: Calibration of canonical model				
Parameter		Value	Source	
CRRA parameter	σ	2.0	Arellano (2008)	
Discount factor	β	0.953	Arellano (2008)	
Risk-free rate	r^*	0.017	Arellano (2008)	
Probability of reentry	ϕ	0.282	Arellano (2008)	
Persistence of z_t	ρ	0.945	Arellano (2008)	
Standard deviation of z_t	σ_z	0.025	Arellano (2008)	
Default penalty parameter	$\delta_z = \delta_{oil}$	0.969	Arellano (2008)	
Low oil rents	R_L^{oil}	0.06	Data for Mexico	
High oil rents	R_H^{oil}	0.08		
Lag between discovery and production	T_{wait}	5	Oil discoveries data	
Probability of discovery	π_{disc}	0.01		
Probability of exhaustion	π_{ex}	0.02		

C Sensitivity to the elasticities of substitution

The benchmark sets the Armington elasticity of substitution to $\mu = 2.9$ in both sectors, following [Mendoza and Yue \(2012\)](#). This appendix repeats the analysis of Subsection 4.3 for a higher elasticity and for elasticities that differ across sectors.

A higher elasticity makes it easier for producers to substitute away from imported intermediates, which reduces the output cost of default. Raising μ therefore works against the mechanism. I set $\mu = 5.1$, the upper end of the range reported by [Bajzik, Havranek, Irsova, and Schwarz \(2020\)](#).⁴⁴ Figure 20 presents the analysis from Figure 8 with $\mu = 5.1$ in both sectors.⁴⁵

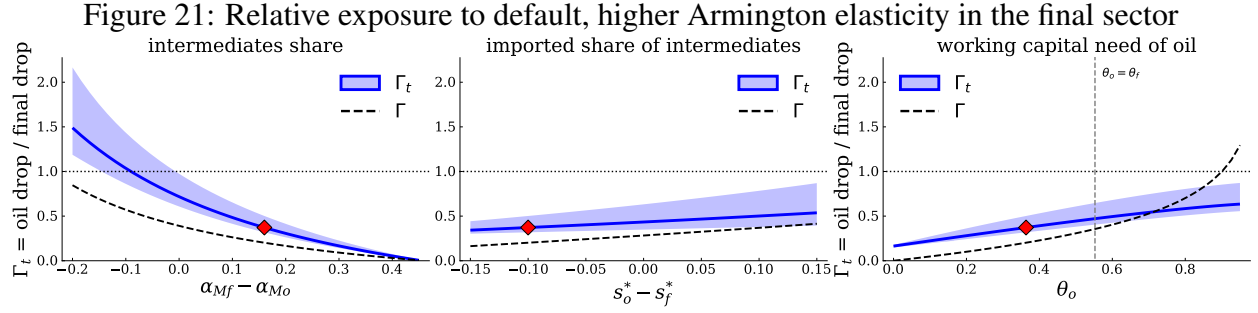


The three panels repeat the top two and the bottom-left panels of Figure 8 with the Armington elasticity set to 5.1 in both sectors. All values are computed from the static equilibrium given the shock and the large oil field. Solid blue lines are Γ_t evaluated at $z_t = 1$ and the shaded areas its range over the grid for z_t . Dashed black lines are the sufficient statistic Γ of equation (33). Red diamonds are the benchmark calibration for Mexico. The bottom-right panel of Figure 8 is omitted because it is computed from measured input-output shares and does not involve the Armington elasticity.

Figure 21 raises the elasticity only in the final sector and keeps the oil sector at its benchmark value. Two considerations support this choice. First, the imported inputs of the oil sector are specialized capital goods and oil-field services, while those of the rest of the economy are a broader set of intermediates with domestic substitutes. [Blonigen and Wilson \(1999\)](#) find that the Armington elasticity is lower in sectors with higher entry barriers and a larger presence of foreign affiliates, both of which describe the oil sector. Second, this choice is also the one that works hardest against the mechanism, because it improves the ability to substitute in the sector that is more exposed to default. In both exercises λ_o is recalibrated at the new elasticity so that the share of domestic intermediates in the oil sector remains at its value in the data.

⁴⁴Recent estimates point in the opposite direction. [Boehm, Levchenko, and Pandalai-Nayar \(2023\)](#) estimate short-run and long-run trade elasticities that are below the benchmark value.

⁴⁵The bottom-right panel in Figure 8 does not depend on the Armington elasticity, so it is omitted.

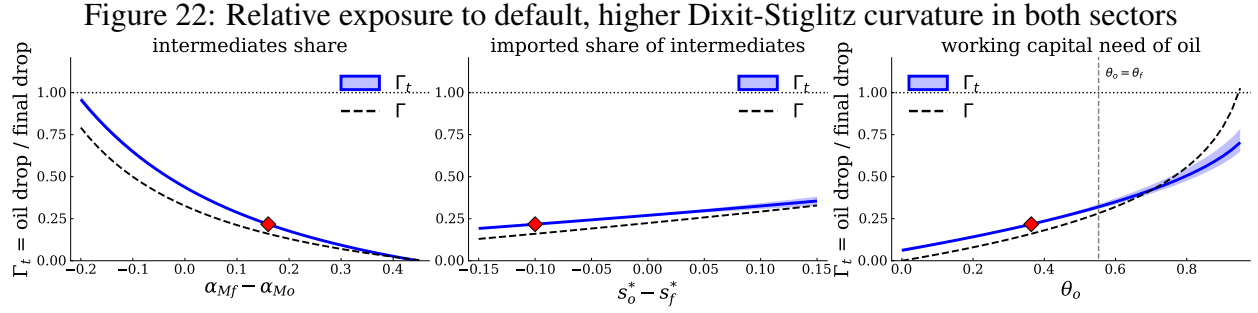


The three panels repeat the top two and the bottom-left panels of Figure 8 with the Armington elasticity set to 5.1 in the final sector and to its benchmark value of 2.9 in the oil sector. All values are computed from the static equilibrium given the shock and the large oil field. Solid blue lines are Γ_t evaluated at $z_t = 1$ and the shaded areas its range over the grid for z_t . Dashed black lines are the sufficient statistic Γ of equation (33). Red diamonds are the benchmark calibration for Mexico. The bottom-right panel of Figure 8 is omitted because it is computed from measured input-output shares and does not involve the Armington elasticity.

The conclusions of Subsection 4.3 do not change in either exercise. The sufficient statistic Γ remains a good proxy for the exact ratio Γ_t , the benchmark calibration for Mexico remains far from the values at which Γ_t reaches one, and Γ_t remains more sensitive to differences in the share of intermediate materials than to differences in the share of imported materials or in the exposure to the financial friction.

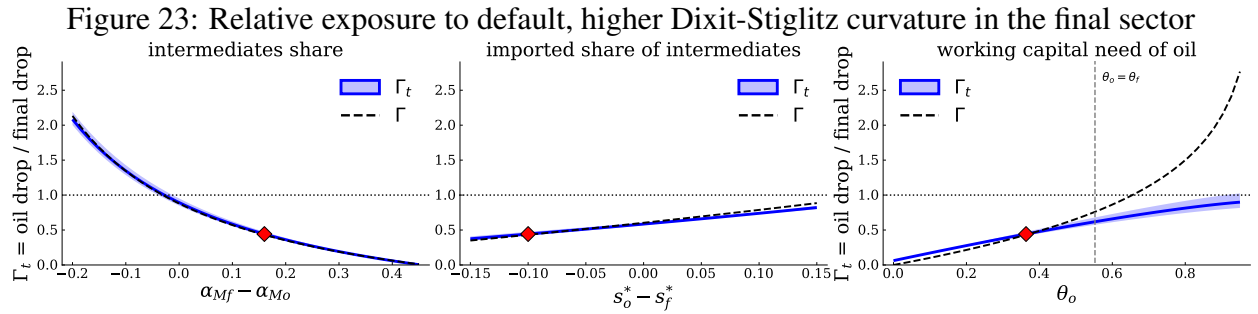
When the two elasticities differ, the range of Γ_t over the grid for z_t is wider than in the benchmark. This is because the sectors now substitute at different rates in response to the same shock, so the ratio of their output drops varies more. This widening is largest where the oil sector uses intermediate materials more intensively than the rest of the economy, far from the benchmark calibration.

The benchmark also sets the Dixit-Stiglitz curvature across imported varieties to $\nu = 2.44$, following Mendoza and Yue (2012). I repeat the two exercises for $\nu = 4.88$, twice the benchmark value. This doubling puts the curvature above the median of 3.1 that Broda and Weinstein (2006) estimate across imported varieties at the ten-digit level and below their mean of 12. Figure 22 raises it in both sectors.



The three panels repeat the top two and the bottom-left panels of Figure 8 with the Dixit-Stiglitz curvature set to 4.88 in both sectors. All values are computed from the static equilibrium given the shock and the large oil field. Solid blue lines are Γ_t evaluated at $z_t = 1$ and the shaded areas its range over the grid for z_t . Dashed black lines are the sufficient statistic Γ of equation (33). Red diamonds are the benchmark calibration for Mexico. The bottom-right panel of Figure 8 is omitted because it is computed from measured input-output shares and does not involve the Dixit-Stiglitz curvature.

Figure 23 raises it only in the final sector, which is again the choice that works hardest against the mechanism.



The three panels repeat the top two and the bottom-left panels of Figure 8 with the Dixit-Stiglitz curvature set to 4.88 in the final sector and to its benchmark value of 2.44 in the oil sector. All values are computed from the static equilibrium given the shock and the large oil field. Solid blue lines are Γ_t evaluated at $z_t = 1$ and the shaded areas its range over the grid for z_t . Dashed black lines are the sufficient statistic Γ of equation (33). Red diamonds are the benchmark calibration for Mexico. The bottom-right panel of Figure 8 is omitted because it is computed from measured input-output shares and does not involve the Dixit-Stiglitz curvature.

The conclusions of Subsection 4.3 do not change in either exercise. The sufficient statistic Γ is in fact invariant to a common value of ν , because the factor $1/(\nu - 1)$ in equations (29) and (30) cancels from their ratio, so only the second exercise changes it.

D Alternative specifications of the model

This appendix uses an alternative version of the model that does not distinguish between aggregate and government consumption $\psi_g = 0$. This version is closer to the canonical sovereign debt model in which the benevolent government makes all choices on behalf of its constituents and has direct

access to the resource constraint. It cannot address the differences in the response of government and private consumption by construction, but it is closer to the data along the dimensions in which the benchmark falls short. To make it comparable to the benchmark, I recalibrate $(\sigma_z, A, \beta, \theta_f)$ following the same SMM approach described in Subsection 4.1.

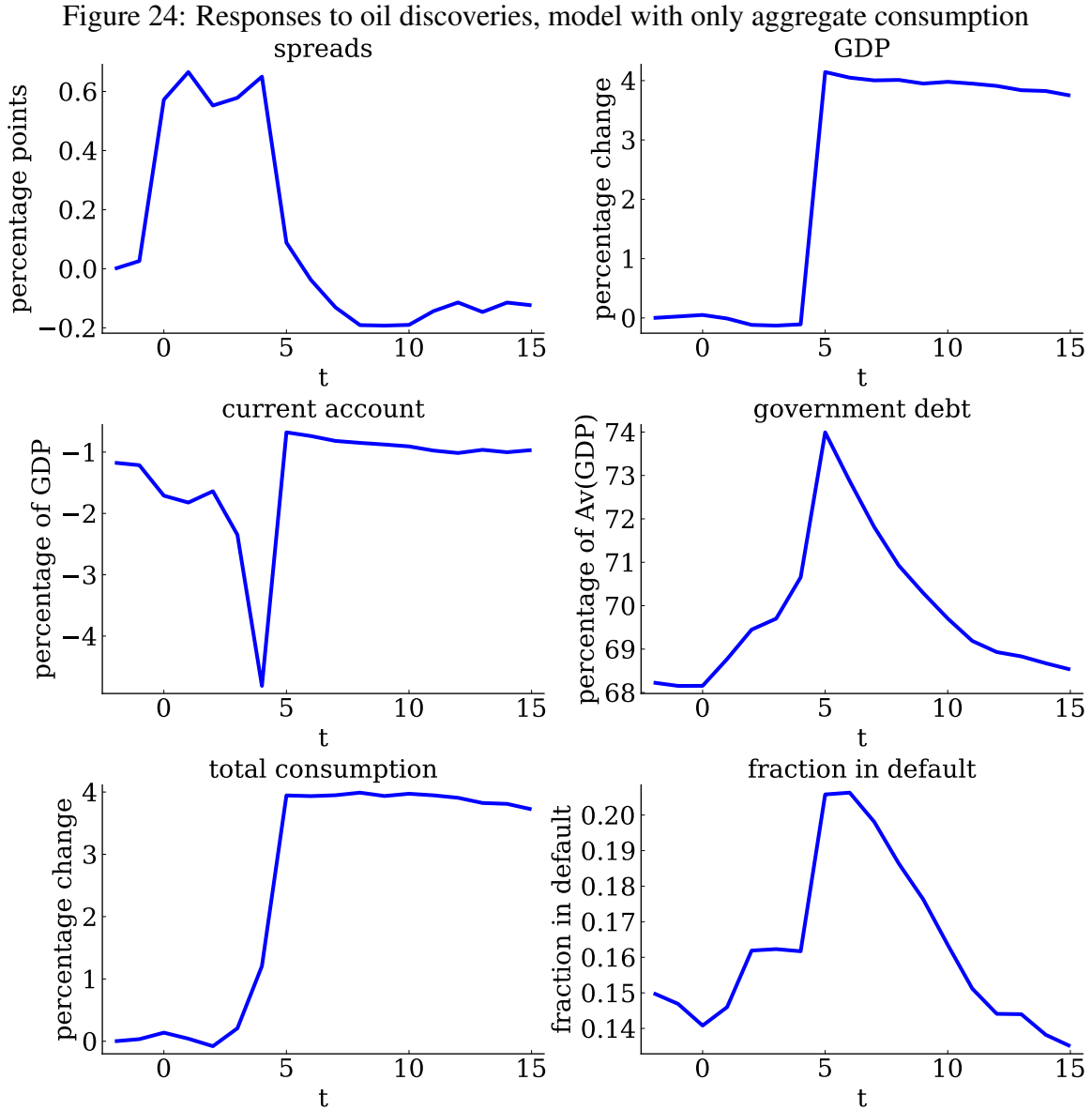
Table 7 reports the business-cycle moments of Table 3 for this model, alongside the data and the benchmark. The long-run default probability and the volatility of spreads are larger than their data counterparts, as they are in the benchmark. The average of the external debt-to-GDP ratio is 0.21 in the data and 0.19 in the benchmark model, and Subsection 4.2 attributes the remaining gap to the relatively small size of the government in the economy and to the assumption that there are no lump-sum taxes or transfers. Consistent with that, the model with only aggregate consumption and a consolidated resource constraint borrows far more, with a debt-to-GDP ratio of 0.64, well above the average of the total-debt-to-GDP ratio in the data of 0.47.⁴⁶ The volatilities of the trade balance and of the current account, which the benchmark understates, also increase substantially: from 0.28 to 1.04 against 2.57 in the data, and from 0.83 to 3.25 against 2.14.

Table 7: Business cycle moments, model with only aggregate consumption

	σ_y	Δy^{def}	μ_{r-r^*}	$\frac{wk}{y}$	$Pr(d=1)$	$\frac{b}{y}$	σ_{r-r^*}	$\frac{\sigma_c}{\sigma_y}$	$\frac{\sigma_g}{\sigma_y}$	$\sigma_{tb/y}$	$\frac{\sigma_{ca}}{y}$	$\rho_{r-r^*,y}$	$\rho_{tb/y,y}$	$\rho_{ca/y,y}$
data	3.11	-13.28	2.90	8.09	2.45	0.21	1.36	1.45	1.05	2.57	2.14	-0.22	-0.19	-0.49
benchmark	3.06	-14.47	2.24	7.94	3.02	0.19	1.87	1.01	1.04	0.28	0.83	-0.25	-0.27	-0.09
no g	3.26	-13.43	2.73	8.10	4.39	0.64	2.22	1.04	n.a.	1.04	3.25	-0.21	-0.21	-0.07

The alternative model is labeled “no g” and is the model with no government consumption. The first four columns are the moments targeted by the calibration of Table 2. Moments are computed by simulating 2,000 time series of 400 periods after trimming the first 100 without any oil discoveries and that are in good financial standing in the initial period. Periods spent in default are excluded when computing the moments. GDP and consumption data are applied the HP Filter with a smoothing parameter of 100. All moments are in percent, except $\frac{b}{y}$ and the ratios of standard deviations and the correlations.

⁴⁶See the Historical Public Debt Database collected by the IMF. The data for the total-debt-to-GDP ratio for Mexico is for the years 1982 to 2015.



Each time series considers 10,000 economies in their ergodic state without any oil discoveries since $t = -50$. The lines are the difference between the values in each period and the values in the first period plotted.

Figure 24 presents the same analysis as Figure 6 for this version of the model. Spreads rise ahead of production and peak in the year before the field comes online, as they do in the benchmark. The paths of output, of the current account and of government debt are also close to those of the benchmark. The differences in technology and working-capital requirements imply a lower default cost in terms of aggregate consumption, just as is the case for government consumption in the benchmark model.